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APS112
Conceptual Design Specifications (CDS)

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Project Title	Improving On-Stage Sound Experience for Dance Performers
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State your word count (excludes Cover Page, Executive Summary, References and Appendices).

Y

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Executive Summary

As reported by our client, Mathura Shivakaran, sound across a performance stage is often projected with either a lack of clarity, synchronization, and or uniformity, with the possibility of all three issues being present at once. Thus, the following document will investigate Hart House Theatre to investigate our client's need for a method of reliable sound projection across a performance stage to help mitigate the acoustic challenges imposed on dancers.

To start, our team conducted a site visit to get a better sense of the problem by understanding first-hand how the problem currently exists and to spark potential ideas for future solutions. During the site visit, the team took many photos, took some baseline measurements of the theatre's response to sound, and documented all aspects of the current state of the theatre.

Using primary data gathered from the site visit paired with secondary research, the team was able to derive the appropriate functions, objectives, and constraints necessary to define what a successful design would look like. More notably, the team was able to specify three key objectives each relating to one of the reported challenges to allow for the measurability of success of our inevitable final design.

Following the creation of our detailed requirements, the team then began focusing on idea generation. Using a number of processes, the team was able to derive 66 different solutions. However, after conducting several more processes the list was soon narrowed down to 3 unique alternative designs; a suspended-surround sound system, an in-ear monitoring system, and a portable acoustic panel and diffuser solution.

The suspended-surround sound system leverages the existing empty ceiling space to project sound in an equally distributed manner. However based on determined estimates, the total cost of the design may exceed our established cost constraint.

The in-ear monitor, dubbed the AuricleSync, completely bypasses the challenges of clarity, synchronicity, and uniformity from speaker based solutions as the sound is projected directly into the dancer's ear. It is also designed to be discreet enough for performances and accessible enough for all ear shapes and types making it suitable for all dancers.

Lastly, the panel and diffuser system, dubbed the AcoustiFlex, takes on a different approach by altering how sound is propagated rather than changing the method of sound projection. It is designed to work in tandem with the existing PA system, is easy to set up, and does not rely on permanent alterations to Hart House Theatre.

Ultimately, after utilizing a Pugh Chart to compare the 3 different solutions, it was determined that the AuricleSync design was the optimal solution based on its advantages in improving audio accuracy, which is the design top objective.

This document marks the end of the team's idea generation, selection, and development. Our client will be expected to review our document to approve or reject our proposed final design in which the team will subsequently start developing our final presentation following her approval.

1.0 Introduction

While practice and effort strongly influence a dancer's performance, the sound environment also plays a crucial role in their success. Poor sound conditions, such as audio delays across the stage, uneven volume levels, and unclear sound quality, can negatively impact a dancer's ability to perform. Mathura Shivakaran, a University of Toronto alumna, has experienced these challenges throughout her dancing career and seeks a solution that projects sound more effectively to address these issues [1].

This document will define the problem and service environment, consider the potential interest holders involved, outline design requirements, describe idea generation processes, and develop alternative solutions to be presented to the client.

2.0 Problem Statement

The client describes performing in poor sound environments where key elements of the music are not clearly projected. Overpowering low-frequency sounds often drown out other musical components and create distracting floor vibrations (Appendix A). These issues make it difficult to identify beats (Appendix A), hindering dancers' ability to synchronize with the music [2] and with other performers on stage. As a result, dancers experience reduced coordination, confidence, and consistency, often requiring spontaneous adjustments during performances [1].

To investigate this problem, the project focuses on Hart House Theatre (Figure 1), a professional performance venue that exhibits similar acoustic challenges to those described by the client (Appendix A).



Figure 1. Hart House Theatre Stage [3]

During a site visit, the team conducted a frequency response test using Spectroid, measuring pink noise picked up by a microphone at a listening position on stage (Figure 2). While an ideal pink-noise frequency response follows a -3 dB slope

[4], the measured response showed pronounced peaks at specific frequencies (e.g., 94 Hz and 155 Hz), indicating modal reinforcement caused by room reflections. These irregularities contribute to uneven, muddy sound on stage [5].

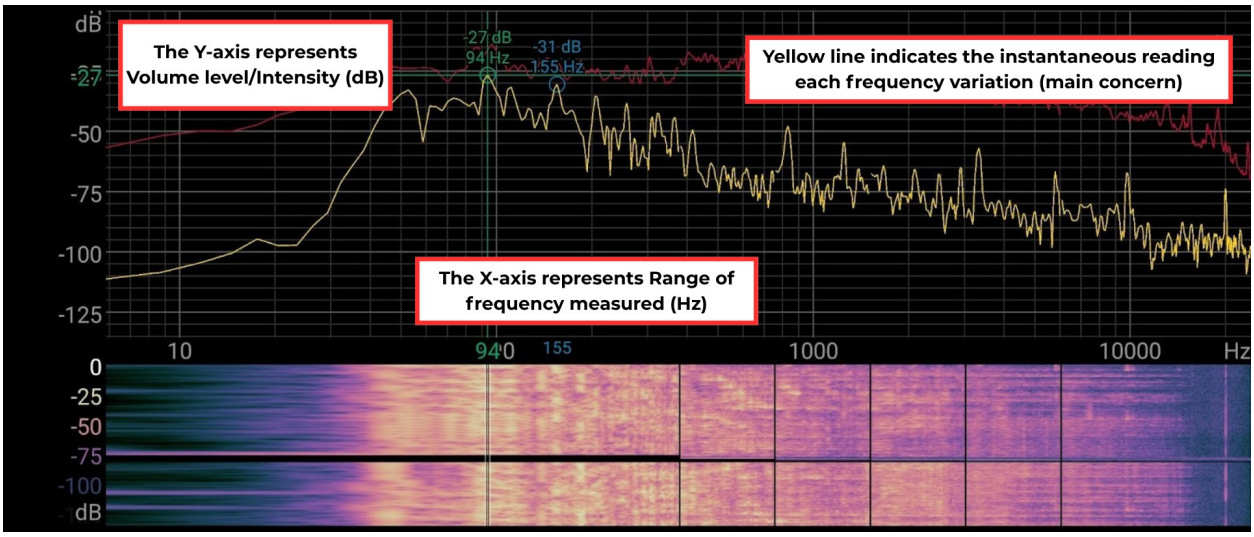


Figure 2. Frequency Response Graph

From this analysis, the gap is that there is a lack of methods to reliably project sound to dancers across a stage. Sound reliability will be defined in terms of clarity, synchronicity, and uniformity across a stage environment: clarity refers to overall audio quality and accuracy, synchronicity refers to consistent timing of perceived sound across performers, and uniformity measures the variance in volume levels across the stage. Related metrics are discussed in sections 5.2 of the report. Accordingly, the need is a solution that ensures accurate, synchronized, and evenly distributed sound reproduction for performers in a stage environment.

3.0 Service Environment

The following sections will discuss three aspects of the service environment, ranked by significance (Appendix B). The physical environment has the greatest impact on our design space. ‘Living Things’ encompasses human factors that affect the design space. Lastly, the ‘Virtual Environment’ section will provide additional details relating to wireless signals.

3.1 Physical Environment

The Hart House Theatre stage is a proscenium stage (Appendix B.2.1), with its dimensions and on stage speakers location shown in Figure 3. Table 1 outlines relevant information regarding the physical environment of the site.

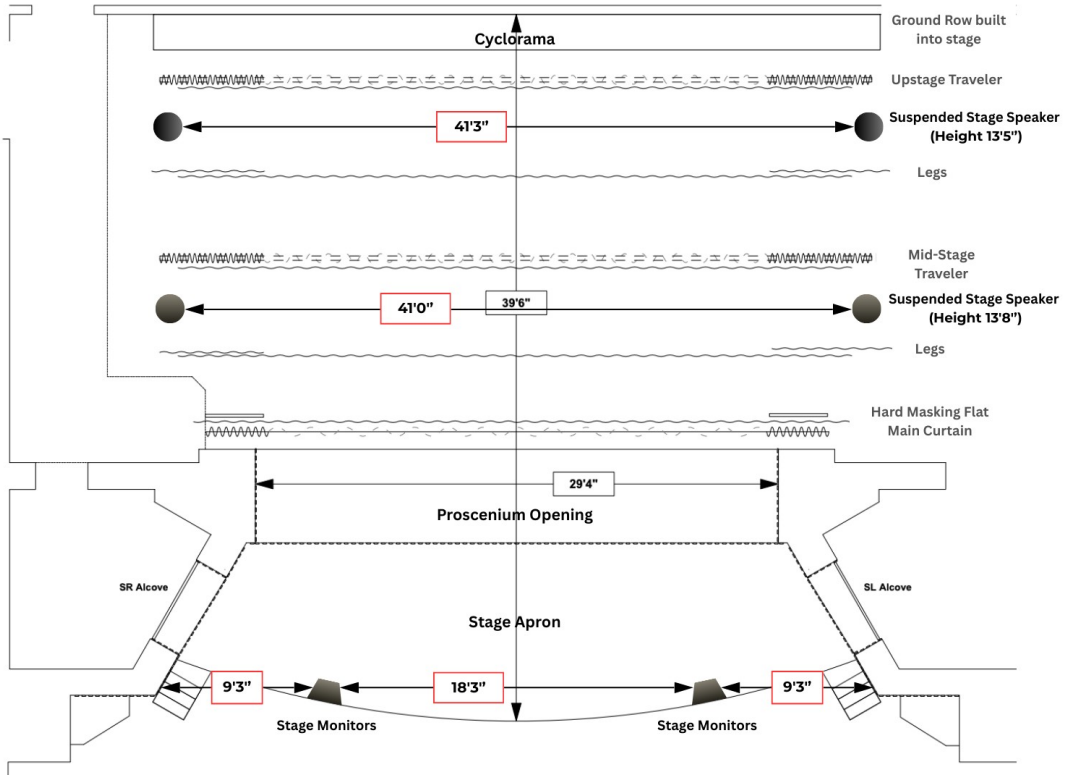


Figure 3. Hart House Theatre Stage Floor Plan and Dimensions (Annotated). [6]

Table 1. Hart House Theatre Physical Environment

Physical Environment		
Stage [7][8]		• Material/Structure: Pine Planking; Soft Pine Wood (Appendix B.2.4) • Flooring: Sprung Floor • Elevated by 1.04m above the first row
Auditorium [8]		• Raked seating ○ 15° inclination • 428 fixed seats
Hardware [8]	AUDIO / PA System	• Control room in a booth elevated above the back of house (Appendix B.2.2) • Audio tech desk in the back corner of the house (Appendix B.2.2), equipped with Yamaha M7CL Digital Mixing Console – 48 Inputs, and two adjustable EQ systems: ○ Graphic EQ system (for speakers) ○ Parametric EQ system (for audio input) • Stage Speakers: ○ Suspended Yorkville YXL12P in concert mode (x4) (Figure 2) (Appendix B.2.5) ○ Stage monitors (x2) (Figure 2) (not used for dance performances)
	STAGE	• Stage is lined with marley (Appendix B.2.3), a vinyl/rubber-

	LINING	like material, during dance performances
	POWER	● 120V @ 15/20A

Additionally, the team conducted audio tests during the site visit to determine the current state of Hart House Theatre's audio system. See Appendix B.3 for details.

3.2 Living Things

As Hart House Theatre regularly hosts various types of shows and performances [3], different types of people will be present within the service environment, as displayed in Table 2:

Table 2. Living Things

Living Things	
● Stage crew: ○ Sound technicians ○ Lighting technicians ○ Set designers ● Maintenance personnel: ○ Venue maintenance technicians ○ Janitors	● Performers: ○ Actors/actress ○ Musicians ○ Dancers

3.3 Virtual Environment

Information related to the virtual environment of the theatre is outlined in Table 3.

Table 3. Hart House Theatre Virtual Environment

Aspect	**Data**
Cellular coverage	3G, 4G, 5G, 5G+, and LTE coverage from most cellular companies [9] [10]
Wifi coverage	UofT and eduroam wifi connections for students, faculty, and guests [11]

4.0 Interest Holders

When designing a solution to improve the on-stage experience of dancers, it is important to consider the various interest holders involved in the process. Table 4 shows the interest holders which have been ranked based on their combination of impact and influence on the project using the Interest Holders Matrix analysis and a pairwise comparison (Appendix C).

Table 4. The Project Interest Holders and Their Impact and Influence

Interest Holder	Impact	Influence

Hart House Theatre Venue Management	**High Impact** As the venue owner and operator, Hart House is responsible for maintaining a high-quality performance environment. Acoustic issues can affect venue reputation [12].	**High Influence** Venue management has authority over theatre policies, equipment investments, and facility modifications.
Hart House Theatre Sound Operators	**High Impact** Audio technicians manage sound mixing and speaker output across the theatre. [13].	**High Influence** Audio technicians control audio related decisions during rehearsals and performances, allowing them to attempt to mitigate acoustic issues in real time.
Hart House Stage Managers	**High Impact** Stage managers coordinate rehearsals and performances. Sound issues can interfere with timing between performers and technical cues, which minimizes coordination during productions for the performers [14].	**Medium Influence** Stage managers coordinate scheduling, stage operations, and communications between performers and technical staff, allowing them to advocate for acoustic adjustments.
Dancers Using Hart House Theatre Student Dance Clubs & Performance Groups at Uoft (e.g. Ballroom Dance Club [15], Hart House Singers [16])	**High Impact** Dancers rely on clear and consistent audio to be synchronized and oriented during the performance. Uneven sound distribution, bass vibrations, and echoes on stage can disrupt their coordination.	**Medium Influence** Dancers cannot make decisions over the theatre, but when experiencing such issues, they can provide feedback to Hart House [17].
Stage Crew of Hart House Theatre	**Medium Impact** Stage crews are responsible for setting up stage elements, speakers, and other technical equipment. Acoustic issues might require troubleshooting before or during performances.	**Medium Influence** Stage crew can adjust equipment placement and stage setup, but are under direction from stage managers and technical staff, hence, limited decision-making authority [14].
Hart House Theatre Audience Members	**High Impact**	**Low Influence**

	Audience members can experience reduced sound quality and performance quality if performers struggle with timing and coordination.	Audience members have limited control over stage acoustics but can influence venue decisions through feedback, attendance levels or reviews [18].
Professional Audio and Stage Equipment Manufacturers	**Low Impact** Manufacturers are indirectly affected by acoustic challenges in performance venues. Persistent issues might highlight limitations of the existing equipment, affecting future sales.	**Medium Influence** Manufacturers influence available technologies and equipment design used in theatres, but are not directly involved with theatre operations [19].

5.0 Detailed Requirements

This section will provide a detailed understanding of the project's core requirements. All functions, objectives and constraints will be scoped to Hart House Theatre stage specified in Section 3.1. The design will primarily focus on optimizing the stage acoustics within this scope.

5.1 Functions

This segment defines the core functions related to audio transmission. The Black Box Method (Appendix D) was used in determining the necessary inputs and outputs of the design [20] (Table 5).

Table 5. Primary and Secondary Functions

Primary	Secondary
Transmit audio from an audio recording to performers on stage at Hart House Theatre.	● Receive electrical audio signals ● Convert electrical audio signals into sound waves ● Transmit sound waves ● Communicate information to users

5.2 Objectives

This section outlines the criteria for achieving reliable sound reproduction, as described in Section 2. These are specified in Table 6 in ranked order, determined via Pairwise Comparison (Appendix E.1).

Table 6. Objectives

Main Objectives	Specified Metric	Goal

Improve Audio Accuracy	Frequency Response (dB)	A flat frequency response (+/- 3dB maximum variation) across the frequency range of 20 Hz to 20 kHz. See Appendix E.2 for a sample graph [21].
Minimize Audio Latency	Milliseconds (ms)	≤10 ms [22].
Volume on Stage	Decibels (dB)	80-85dB [23].
Affordable	$CAD	≤$1000 (Appendix A)
Ease of Use	Steps to operate	≤3 steps [24].

5.3 Constraints

This section defines several requirements to ensure the feasibility of the design with respect to safety, budget requirements, and an effective sound transmission system. These constraints are based on the client statement, existing guidelines, and the service environment and are presented in Table 7.

Table 7. Design Constraints

Description	Specified Metric	Limit
Affordability	$CAD	≤$5000 (Appendix A)
Volume Safety	Decibels (dB)	≤97dB [25].
Accuracy of Audio	Frequency Response (dB)	A flat frequency response (+/- 5dB maximum variation) across 20 Hz to 20 kHz. [26]
Electrical Safety	Electrical parts left unguarded	0 parts [27]

6.0 Idea Generation, Selection, and Description of Alternative Designs

Following the problem definition, our team was able to come up with 66 different solutions spanning multiple themes of the design space of the project. The following sections explore our processes related to the generation and selection process for our three alternative designs, and their descriptions.

6.1 Idea Generation Process

From the 66 derived solutions, the team was able to divide the design space into 6 different sub-spaces, which are all possible approaches in addressing the problem as expressed in Section 2. These are:

- Acoustic/Architectural Treatment
- Loudspeaker System Design
- Proximity Focused
- Beat/Tactile Focused
- Concentrated Audio
- Visual Cues

The following figure demonstrates the process for idea generation and how we were able to derive our 66 solution set. (Figure 4)

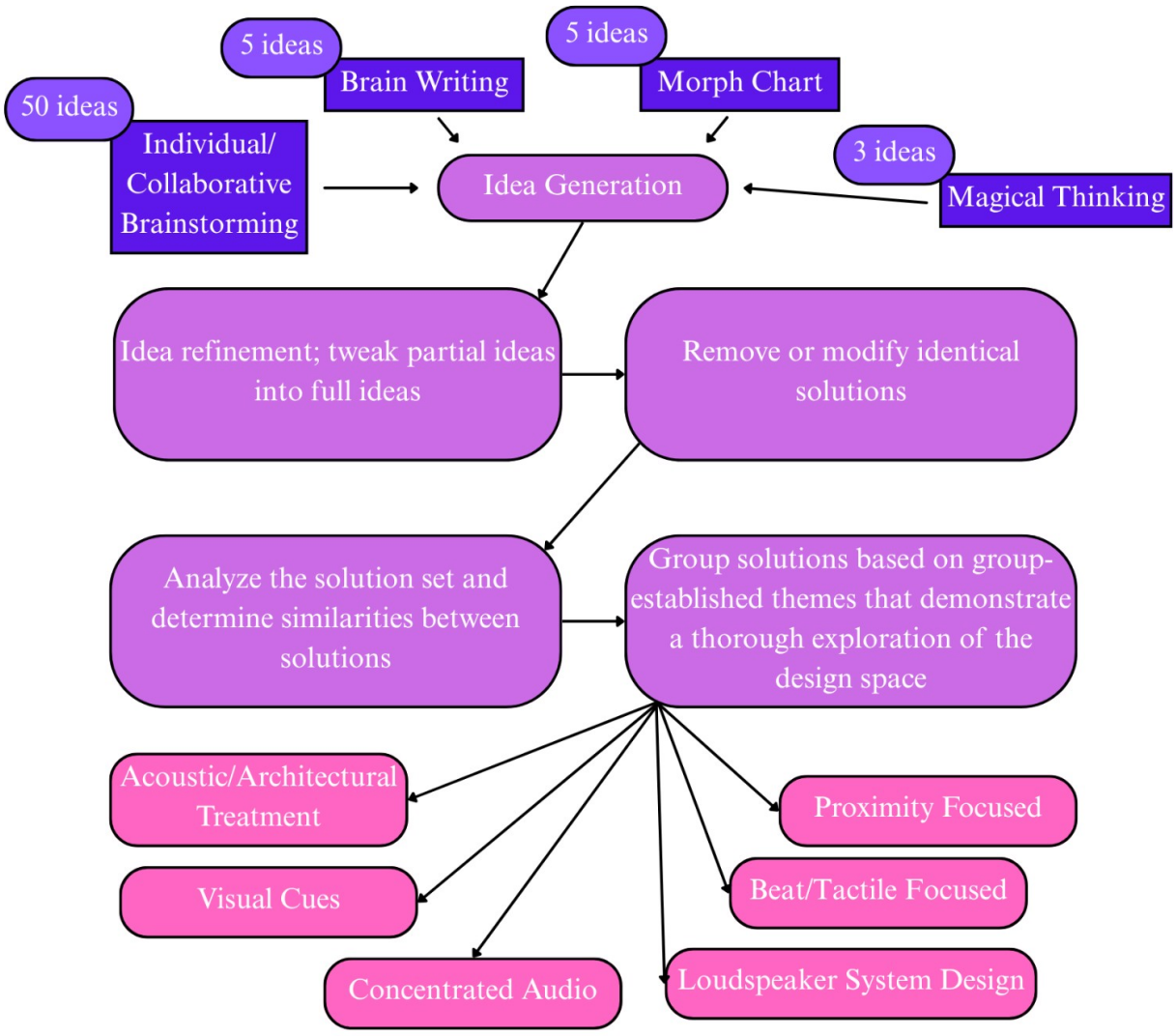


Figure 4. Idea Generation Process Visualization

Individual brainstorming was the only task completed independently, with each team member presenting their own ideas. All other stages of the process were conducted collaboratively. (Appendix F).

6.2 Alternative Design Selection Process

Following the idea generation process, the team then began to narrow down and eliminate our solutions to obtain our final three alternative designs. The process is outlined below in Table 8**.**

Table 8. Alternative Design Selection Process

Original number of solutions		66
Method	**Description**	**Total number of remaining solutions after method**
Consolidation	Remove/combine duplicates	60
Feasibility Check	Remove solutions that are unrealistic or could not be modified to meet constraints	39
Multivoting	Remove solutions that received no votes	12
Consolidation	Apply SCAMPER method to remaining solutions	8
Graphical Decision Chart	Compare solutions against the top two objectives and choose the best three	3

Refer to Appendix G for more details on the method used and the data before and after utilizing each method.

6.3 Alternative Design Descriptions

After narrowing down the solution set to 3 distinct solutions, the following solutions remained:

Immersed-On-Stage: Suspended Surround Sound System	AuricleSync: Integrated Costume In-Ear Monitoring System	AcoustiFlex: Portable Acoustic Panel and Diffuser System

The following sections provide a more in-depth overview of each design, complete with a figure, an explanation on how it will be implemented, and how it will satisfy the established FOCs.

6.3.1 Design 1 - Immersed-On-Stage: Suspended Surround Sound System

Immersed-On-Stage proposes a 5.1 surround sound speaker system (5 speakers) suspended from the stage ceiling using metal struts for a rigid connection. The design receives audio and transmits it from each speaker synchronously. To communicate operation instructions, the design includes a link to an online video tutorial walkthrough and professional help will be required for safe installation. See Figure 5 for speaker layout and stage dimensions.

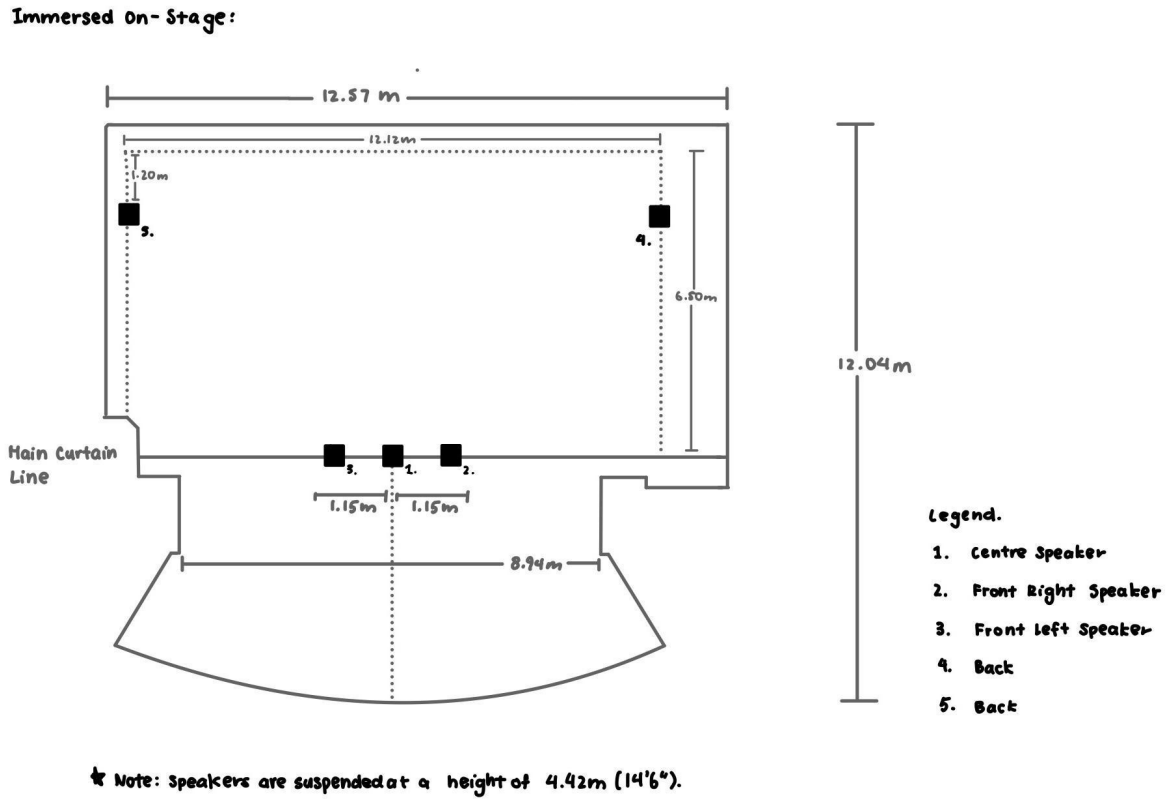


Figure 5. Speaker layout and dimensions

The system uses spacing and angle calculations to optimize audio accuracy through minimization of destructive interference (Appendix H) that results in uneven frequency response readings. The measurements shown in the figure are based on the current stage layout as per Service Environment section 3.1. See Appendix H for calculations regarding spacing and angle of speakers. Suspended speakers offer a greater freedom in placement. The design leverages the ceiling space to ensure speakers are distributed and are in equally close proximity (14'6" in height) to all dancers. Table 9 assesses AuricleSync against objectives, while constraints assessment is in Appendix J.

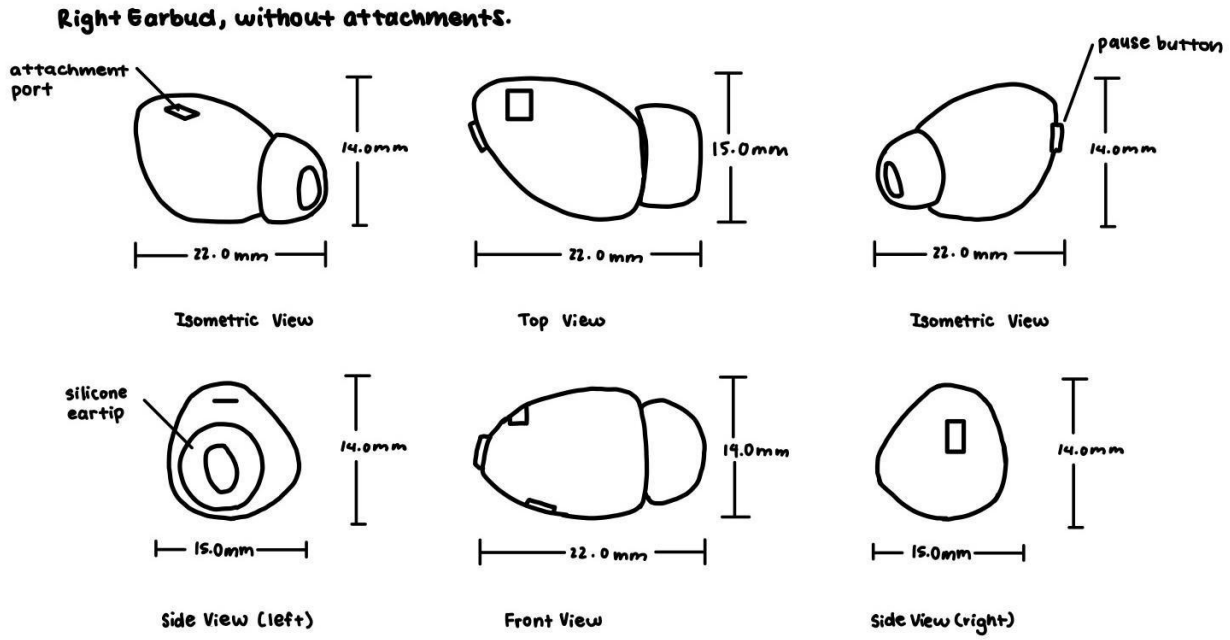
Table 9. Immersed-On-Stage Objective Assessment

Objective	Assessment
Improve Audio Accuracy	Reduces destructive interference and uneven frequency response by ensuring a seamless blend between speakers and correct speaker alignment.
Minimize Audio Latency	By positioning speakers closer and equidistant to performers, the design reduces propagation delay differences across the stage.
Volume on Stage	Even sound distribution reduces the need for excessive amplification, allowing volume to remain within the ideal 80-85 dB range while

	maintaining clarity.
Affordable	The use of multiple professional-grade speakers and structural installation (ie. mounting) increases costs, estimating costs at around $4968.40 (Appendix H).
Ease of Use	The system requires professional installation and calibration, increasing setup complexity. While operation post-installation is straightforward, initial implementation exceeds the ≤3-step usability target.

6.3.2 Design 2 - AuricleSync: Integrated Costume In-Ear Monitoring System

The AuricleSync proposes an integrated in-ear monitoring system embedded within the dancers' jewellery or costume elements near the ear. Figure 6 describes the dimensions of a proposed AuricleSync. This system discreetly delivers high-fidelity audio directly to each performer, enabling accurate synchronization with music while maintaining the visual integrity of the performance and safety of the performers.



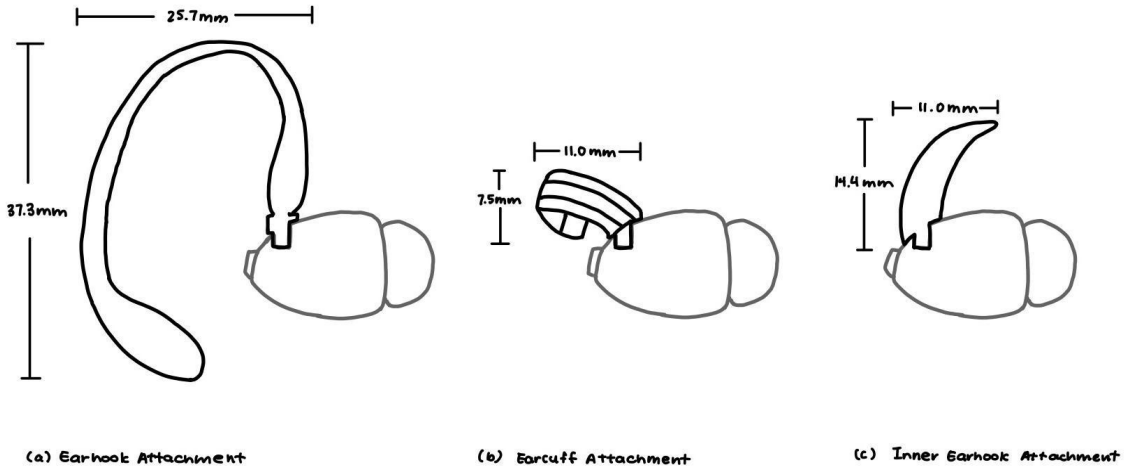


Figure 6. AuricleSync dimensions

During a performance, the modified jewellery piece is worn around the ear. To minimize interference with any accessories worn by dancers, AuricleSync employs different attachments suited to the performer's needs. The device houses a miniaturised speaker positioned near the ear canal, allowing audio to be transmitted directly to the performer. Each unit wirelessly receives signals from a central audio transmitter located off-stage, ensuring dancers receive synchronised audio cues in real time without relying on stage speakers. Additionally, to maximize audio accuracy, the design features noise cancellation. Table 10 assesses AuricleSync against objectives, while constraints assessment is in Appendix J.

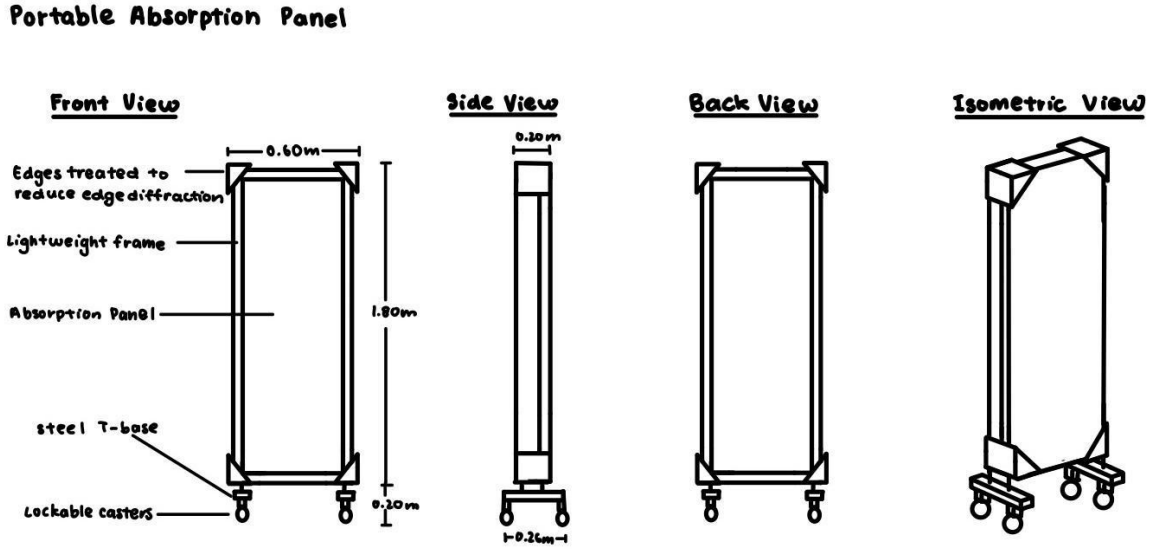
Table 10. AuricleSync Objective Assessment

Objective	Assessment
Improve Audio Accuracy	AuricleSync delivers sound directly to the performer's ear ensuring consistent audio quality regardless of position on stage.
Minimize Audio Latency	Wireless transmission technologies with low-latency protocols are used to ensure near real-time delivery.
Volume on Stage	Transmitting audio directly to performers minimizes reliance on synchronising high-volume stage speakers, which lowers overall stage noise levels, improving sound clarity, and reducing auditory strain for dancers.
Affordable	The system leverages compact, commercially available audio components (e.g. micro-speakers and wireless receivers) on existing jewellery, allowing for scalable production at relatively low cost compared to large-scale professional monitoring systems.
Ease of Use	Minimal setup is required, as performers simply wear the accessory and

	switch on the device to connect to the central transmitter prior to the performance.

6.3.3 Design 3 - AcoustiFlex: Portable Acoustic Panel and Diffuser System

AcoustiFlex proposes a modular system of portable acoustic panels and diffusers that work alongside the existing Public Address (PA) system at the theatre. Instead of modifying the audio signal electronically, the design improves sound quality by controlling how sound waves propagate within the stage environment. See Figure 7 for detailed visuals of the design and refer to Appendix I for a detailed design description.



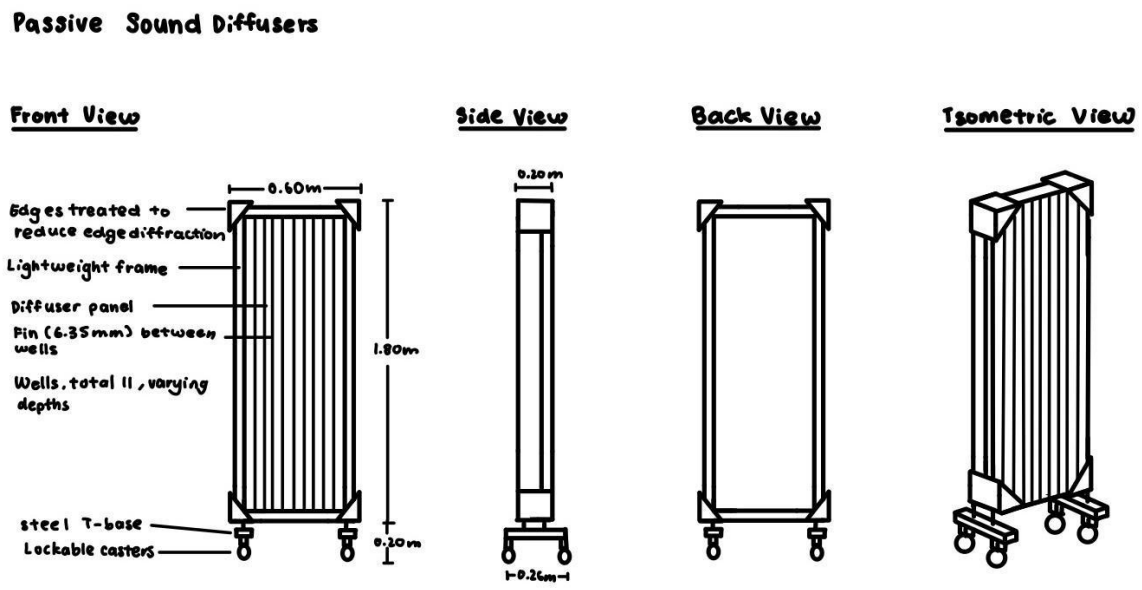


Figure 7. Conceptual AcoustiFlex Design

During implementation, stage crew transport the panels onto the stage and position them strategically based on speaker layout and stage geometry (Appendix I), which requires no permanent installation or modification to existing theatre infrastructure. The absorption panels attenuate reflected sound energy, reducing echoes and reverberation [28], while the diffusers scatter incoming sound waves to prevent strong, direct reflections toward performers [29]. Together, these treatments mitigate reverberation and standing-wave effects, which are key contributors to uneven frequency response and sound "muddyness" [30]. Assessments of AcoustiFlex against each objective are presented in Table 11, while constraint assessment is in Appendix J.

Table 11. AcoustiFlex Objective Assessment

Objective	Assessment
Improve Audio Accuracy	AcoustiFlex reduces resonance and frequency build-ups by scattering sound waves, preventing standing waves, and minimizing deviations from the original audio signal through proper panel placement.
Minimize Audio Latency	Have no direct contribution to electronic delay (dependent on PA system and positioning), although the panels reduce reverberation that worsens audio intelligibility and dancers' temporal perceptions [31]
Volume on Stage	Better audio intelligibility reduces the need to push volume levels to ensure audibility, hence, easier to maintain 80-85dB while preserving clarity.
Affordable	Benchmarking: GIK Acoustics markets a 60x120cm acoustic panel for

	$152 [32], and a 50x117cm diffuser for $626 [33], implying challenges to manufacture the desired quantity (Appendix I) for under $1000.
Ease of Use	Exceeds the ≤3-step target: 1. Identify panels placement 2. Roll/move panels 3. Set angle 4. Lock casters

7.0 Proposed Conceptual Design Specification

To decide the final proposed conceptual design, the team utilized the Pugh Chart tool (Appendix K), where a 5-point scale system was used to compare solutions with respect to the objectives.

Taking into account the three criteria established for sound reliability (Section 2), the service environment, and interest holders, AuricleSync is considered to be the most effective in addressing the client's needs. AuricleSync best satisfies the key criteria of audio accuracy, latency, and volume consistency by delivering sound directly to dancers' ears. This eliminates the effects of environmental acoustics, such as reflections and uneven sound distribution, ensuring synchronized and uniform audio across the stage. In comparison, Immersed-On-Stage improves sound distribution but remains limited by acoustic interference and high installation costs, while AcoustiFlex reduces reflections but does not directly address latency or ensure consistent sound delivery.

AuricleSync is also highly compatible with the service environment, requiring no permanent modifications, and aligns well with interest holders by improving performer experience while minimizing setup complexity for stage crew. Although considerations such as comfort, safety, and wireless reliability exist, these can be mitigated through proper design. Overall, AuricleSync provides the most effective and practical solution for the identified problem.

8.0 Measure of Success

The Measure of Success will act as a road map to measuring the effectiveness of our proposed solution based on the objective, "Improve Audio Accuracy" as per section 5.2.

The chosen objective encompasses the most important criteria relating to effective sound reproduction, ensuring that performers hear the exact same audio as the original. This includes reproducing the same beat and relative frequency volumes that performers are currently accustomed to.

Please see Table 12 for the plan for acquiring the materials needed to build and test the prototype.

Table 12. Materials/Tools needed

Tools Needed	Status	Price
Computer/Laptop	Use available team member's laptop	N/A
Measurement Software (Room EQ Wizard)	Available to download (free)	N/A
AuricleSync Prototype	Beats Solo Buds earpiece: Online purchase	$79.99 [34]
	Floral Bollywood-style Earring: Online purchase	$28.71 [35]
Audio Interface	Borrowed from Hart House	N/A
Microphone (flat frequency response mic to ensure the microphone quality won't alter test results)	Borrowed from Hart House	N/A
Ear Model (Tube simulating ear canal)	Online Purchase	$21.90 [36]
Cables to connect audio interface to computer, and audio interface to device	Borrowed from Hart House	N/A
Total Cost		$130.60

The proposed plan is a frequency response test, a modified version of the test used for speakers tailored to our earpiece-based design. The following image (Figure 8) and plan (Figure 9) are based on the Room EQ Wizard tutorial by Audio University. See video tutorial for a more detailed outline of software and equipment set-up. [37]



Figure 8. Connections and microphone set-up.

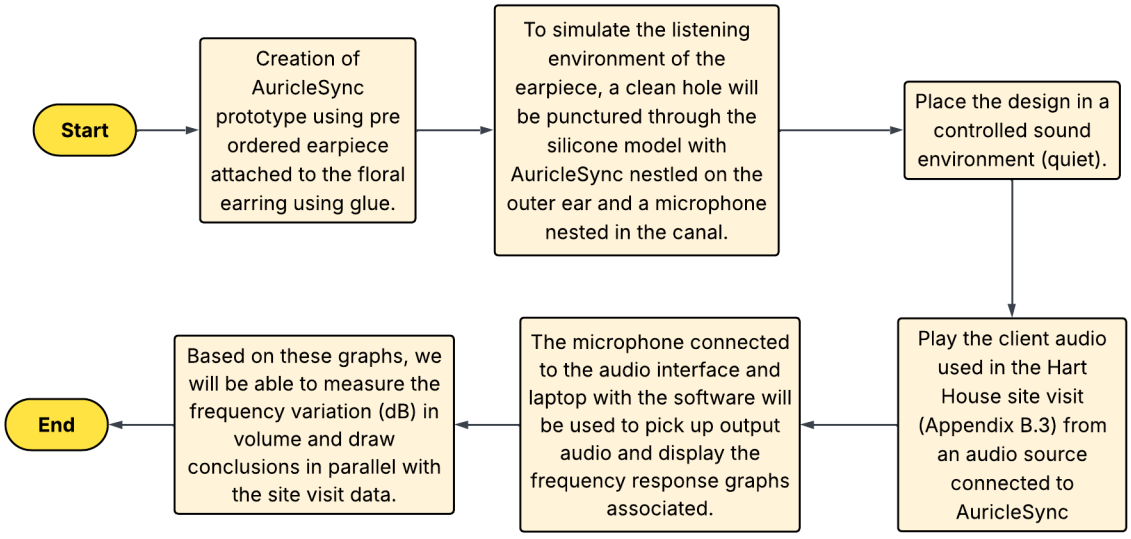


Figure 9. Proposed testing plan.

To ensure the timely completion of the MoS testing phase, the following timeline shows the plan for data collection and analysis relative to the final presentation deadline (Figure 10).

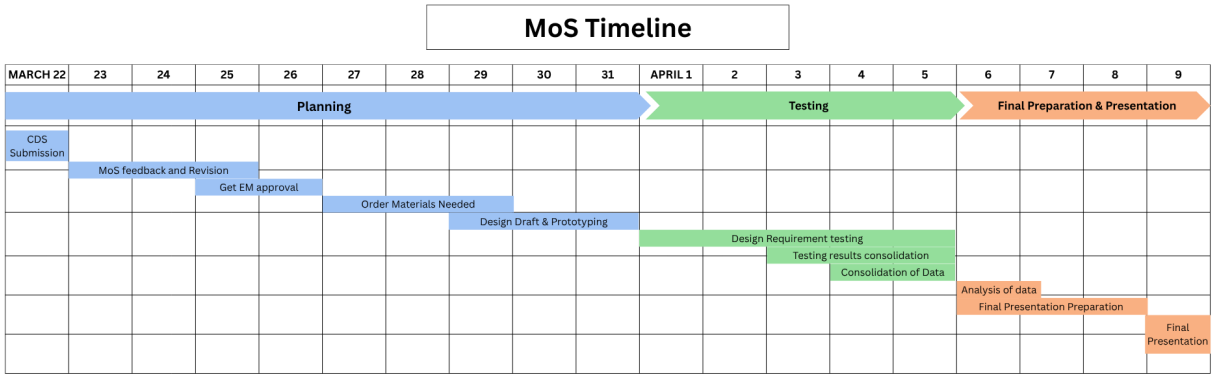


Figure 10. MoS Timeline

9.0 Conclusion

This document presented a comprehensive design process to improve the on-stage sound experience for dance performers, beginning with the analysis on the service environment at Hart House Theatre, identification of key interest holders, and the detailed functional requirements, constraints, and objectives of a proposed solution, before progressing through conceptual design and evaluation of the measures of success.

Through structured ideation and systematic selection methods, 66 initial concepts were refined to three final designs. From these, AuricleSync was selected as the most suitable solution based on its alignment with key project objectives within the given constraints.

The proposed measure of success plan further ensures that the design can be quantitatively evaluated against performance targets such as audio clarity, synchronization, and accessibility. Overall, this document provides a well-supported and practical design direction, forming a solid basis for future development and implementation.

10.0 References

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Appendix A: Client Meeting 1 Notes

This appendix includes questions and notes taken during the Client Meeting 1 with Mathura Shivakaran that was conducted on January 29, 2026, from 6 to 7 pm.

OTHER PRIOR IMPORTANT NOTES:

- Sound distribution is different between
- Performed at stages, the ground, outside
    - Biggest issue is prevalent on stages
    - Same with others
- Affects coordination, confidence,
- Could also be psychological ("I could miss a beat")
- Very experienced dancer (20 years?)
- Specific type of dance to focus on
    - Wants to be accessible and applicable for as many **DANCERS** as possible
    - (REPEATED MOTIF OF **ACCESSIBILITY**)
- The music that she usually dances too is based on a lot of rigid movements
    - Feels more powerful
    - Other styles of music with more melodious types of music may not be the same
    - Bass is important
    - Should account for bass AND melodious music
- Vibrations in the ground usually **throw off** the dancers
    - MINIMIZE VIBRATIONS
- When using speakers, usually only the high notes (high fq) are heard but in performance environments the sound gets more BUSY and MIXED
- Dancers sometimes miss first beat
- Hearing the echoes coming off from the performer themselves might pose a problem
    - Ie drums or piano echos
    - Ie hollow stages (vibrations)
- Most overpowering notes when on STAGE is the BASS
- A lot of performances usually happen in school auditoriums
- KEEP IT INDOOR (solutions)
- Other dancer stomps and steps can also throw off performers
- STAKEHOLDERS
    - Dancers
    - Stage performers
    - Audio techs
    - Stage managers
    - Venue owners
    - Event organizers /
    - Audience
    - Equipment manufacturers
- CONSTRAINTS
    - Safety and building regulations
        - MUST BE SAFE TO USE???
    - Should not interfere with stage lighting or other preexisting stage control systems
    - Should be able to function in environments with lots of noise
        - A lot of shaking or high / low frequencies
    - Not too much electrical interference
- Consider vibration studies and how sound differs

1. What is your motivation behind this project? Is this project intended to be conducted independently or in collaboration with the TTC? **(JING)**

- No correlation to TTC
- Has participated in some teaching opportunities
    - Has had some kids complain about this
    - Younger kids
- No company association
- Probably might include some uni hs
- She teaches at a dance studio
    - 4-23 year old students

2. Is there a specific site/stage that you think we should focus on? *Considering your background as a dancer, have you ever performed on a stage where you experienced the challenges that you mentioned?* **(ANDRE)**
    - Would ideally like to keep it broad and adaptable to multiple stage environments
    - Could use Hart House Theatre for testing or benchmarking??

    a. What challenge would you say affects you the most and why?
        - Lack of a stage rehearsal
        - Minimal time to practice on the stage of the performance
            - The way sound bounces back to the stage
            - Not all stages or venues offer this luxury
            - Rehearsals typically use small phone speakers
        - Certain bass sounds might sound more amplified depending on where you are on the stage
            - Could also be stage features
                - Wall material
                - Wall shape
                - Ground material
        - When closer to the speakers, some sounds might be too amplified
        - Stage settings that the sound crew uses may also be a factor
3. What type of stage performer should we tailor our solution to? **(LINA)**
    - Dance performers is acceptable
    - Keep in mind the accessibility aspect of the design
    - Could maybe broaden the scope to other performers

    a. What kind of problems are specific to this particular stage performer?

    b. Types of dance, how many performers, live music or audio, external noise, volume range, information sheet available? (Service environment)
        - Some experience with live music
            - Other issues that come with live music
        - Focus more on the speakers bc that is mostly the main issue
        - Roughly 5-12 people on stage at the same time
            - Varies
            - 20 max
        - The way the volume comes out
            - Could look at production level speakers
            - Could talk to Hart House for performance level speakers

4. Can you compare/describe in more detail the sound-related challenges you experience during actual stage performances and rehearsals? **(NAUFAL)**
    a. How exactly does it impact their performance or experience? (Function)
    b. How big is the difference between rehearsals and performances?
        - Vibrations
        - Echoes
        - Delay from when the beat comes out on the speaker and when the performers hear the beat
            - If one person is thrown off, the whole group might be thrown off too based on visual cues
5. What is the budget or the scale of the project and are there any limitations in terms of resources, maintenance period, material used or general set-up? (Constraints, interest holders) **(LINA)**
    - Max of $5000 LIMIT
    - Ideally under $1000 OBJECTIVE
    - Affordability
        - Prioritize the enjoyment aspect
    - Accessibility
        - For smaller dance teams
        - Accessible for all
    - Max 20 items if design is for each individual.
    - If wearable devices, consider sizing, ease of use, safe, doesn't interfere with the dance, repeated uses, portable??? (applicable for most settings),
6. What, in your opinion, should be the biggest priority / biggest objective for the project when comparing rehearsal vs performance environments? **(CHLOE)**
    - Making sure that the way sound is delivered or felt is as similar as possible
7. In your opinion as a dancer, what aspects of the music would you consider to be vital in aiding performance? (Objectives) **(NAUFAL)**
    - Beats are one of the **most important**
        - No beat = no dance
    - Other aspects of the music comes after; times steps according to
        - Other instruments
        - Specific phases
        - Specific lyrics
    - BEATS are really important but for cohesiveness and for full performances other aspects are also important
8. Have you ever tried any solutions? Did they work or not? Why? **(CHLOE)**
    - Have not tried any wearable solutions bc there is a lot of jewelries involved with her specific type of dancing
    - Mostly have just been toughing it out
        - Her dance teacher does not like using technology
        - Price-wise its not worth it
    - Toughing it out:
        - As you grow older and gain more experience it becomes more noticeable
        - It takes time to adjust

        - The start of the performance is typically when most mistakes are made
    - Learn to use other people as a guide as well (VISUAL)
    - On the fly adaptation

a. What are some current solutions that help bridge the difference between rehearsals and performances?

9. Why do performers often not have the luxury of stage rehearsals? **(ANDRE)**
    - No stage crews
    - Depends on event organizier
        - The bigger the org the more professional
        - Even if u ask they wont always give it
        - Smaller groups might not be as prestigious or worthy enough to sacrifice resources for stage rehearsals
            - Ties to **accessibility**
                - Everyone on stage should be given an equal and fair chance
                - Children can get really discouraged

a. What are some limitations that prevent them from having regular stage rehearsals?

10. What success criteria should be prioritized? (Objectives) Examples: **(DARIAN)**

a. Affordability
b. Ease of set-up
c. Accuracy of beat
d. Making stage rehearsals accessible
e. Even sound distribution

- Improved timing accuracy
    - Minimize delay
- Ease of set up
- User friendly (not too complicated to use or implement)
- Affordability (least important)
- Time of set up (ie wearable devices)
    - Could depend on type of performance but for dancers doesn't really matter bc same costume or fit
- Does not interfere with clothes or costumes

Weight (ideally minimized)

Appendix B: Service Environment

This appendix section will include all elements related to the service environment, including relevant rankings and pictures from the site visit.

Appendix B.1: Service Environment Ranking

Table B.1. shows the Pairwise Comparison to determine the importance of each service environment section.

Table B.1. Service Environment Pairwise Comparison

	Physical Environment	Living Things	Virtual Environment	**Score**	**Rank**
Physical Environment		1	1	2	**1**
Living Things	0		1	1	**2**
Virtual Environment	0	0		0	**3**

Appendix B.2: Site Visit Photos

The team conducted a site visit to Hart House Theatre on March 13, 2026 and took several pictures to better capture the current state of the theatre, which are shown in Figures B.2.1 to B.2.5.



Figure B.2.1. Hart House Stage (audience perspective)



Figure B.2.2. Annotated view from stage



Figure B.2.3. Rolls of floor lining (marley) used for dance performances



Figure B.2.4. Close-up of stage floor (without marley)

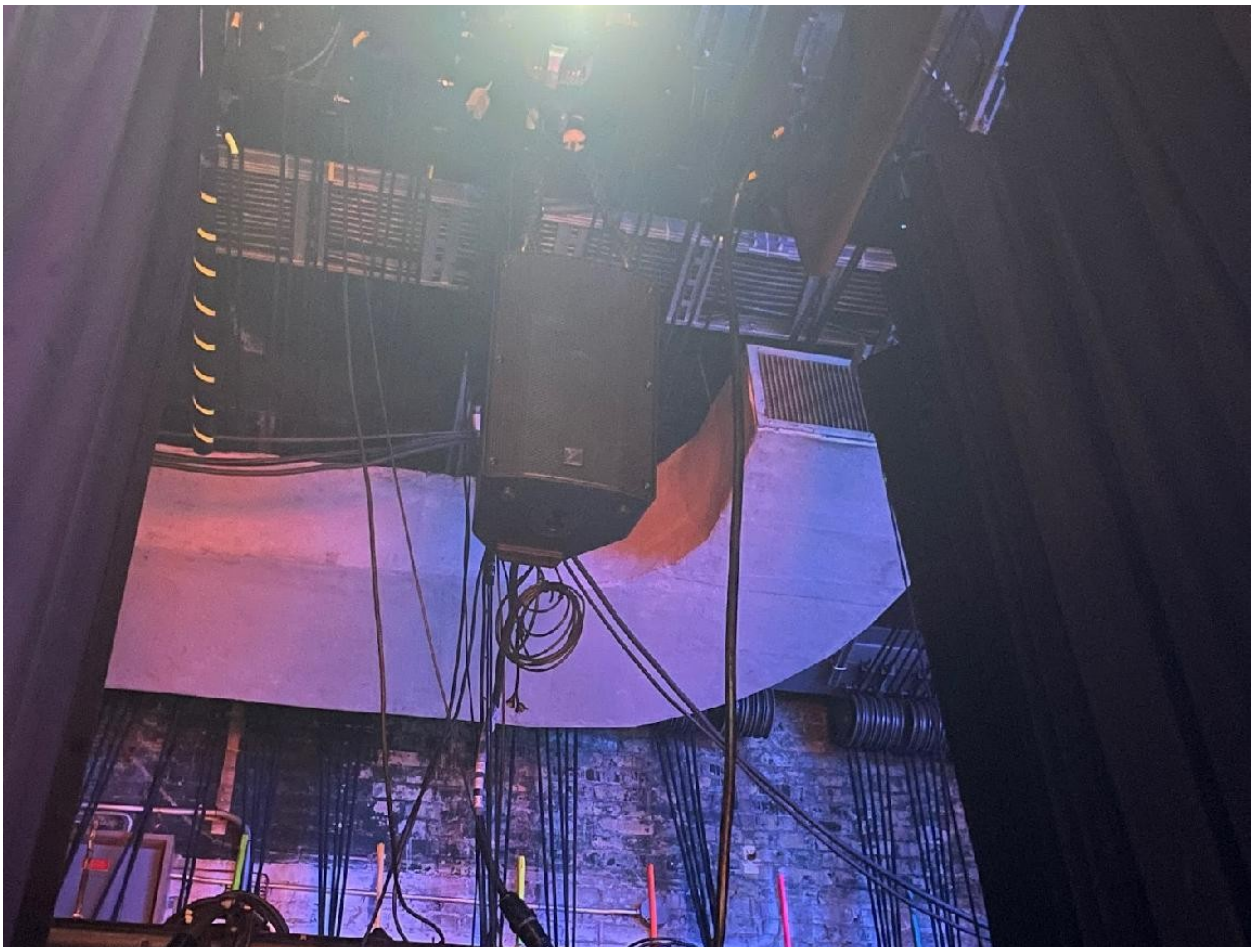


Figure B.2.5. Suspended Yorkville YXL12P stage speaker (1 of 4)

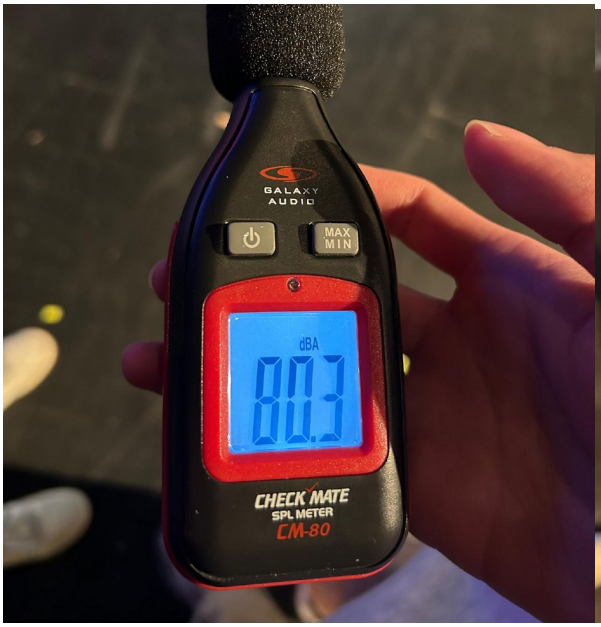
Appendix B.3: Site Visit Audio Test Data

The following figures show the frequency response data from different areas on stage at Hart House theatre. Sample audio from the client [38] was used to produce this data and determine the problematic frequencies in question. See the following link for the sample song used: Please see the Objectives section 5.2 for insight into the preferred results for a frequency response test. White dots reflect the true value, while purple bars are instantaneous. Use white dots as a reference for measurements.



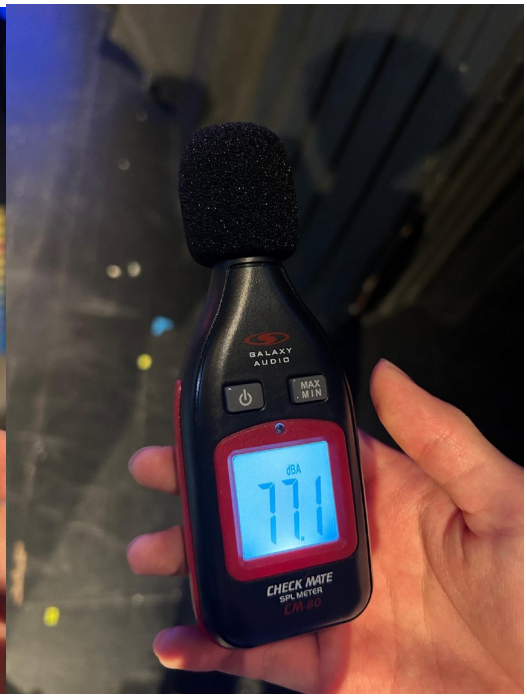
Figures B.3.1. Frequency Response Graphs

The following figures show an audio test using a single 1kHz tone projected by the speakers, measuring volume in decibels at different locations in the theatre. The order of locations (from left to right and top to bottom) is Front Right, Front Center, Center Stage, Front Right, Back Center, Back Right and Back Left.


GALAXY AUDIO
MAX MIN
dBA
80.3
CHECK MATE
SPL METER
CM-80

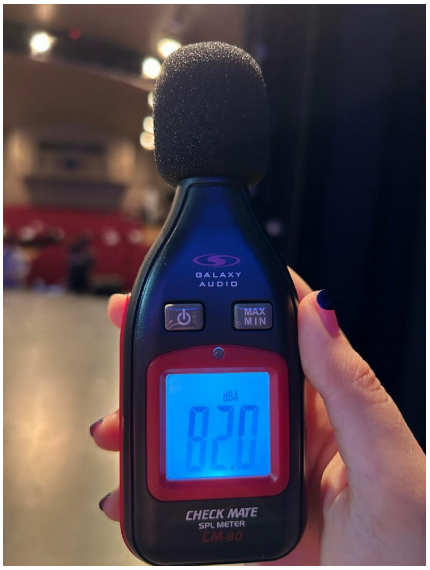

GALAXY AUDIO
MAX MIN
dBA
73.0
CHECK MATE
SPL METER
CM-80


GALAXY AUDIO
MAX MIN
dBA
82.2
CHECK MATE
SPL METER
CM-80


GALAXY AUDIO
MAX MIN
dBA
77.1
CHECK MATE
SPL METER
CM-80







Figures B.3 - 5-11 Volume on Stage Audio Test

Appendix B.4: Hart House Site Visit Additional Notes

This section contains handwritten notes taken during the site visit to Hart House.

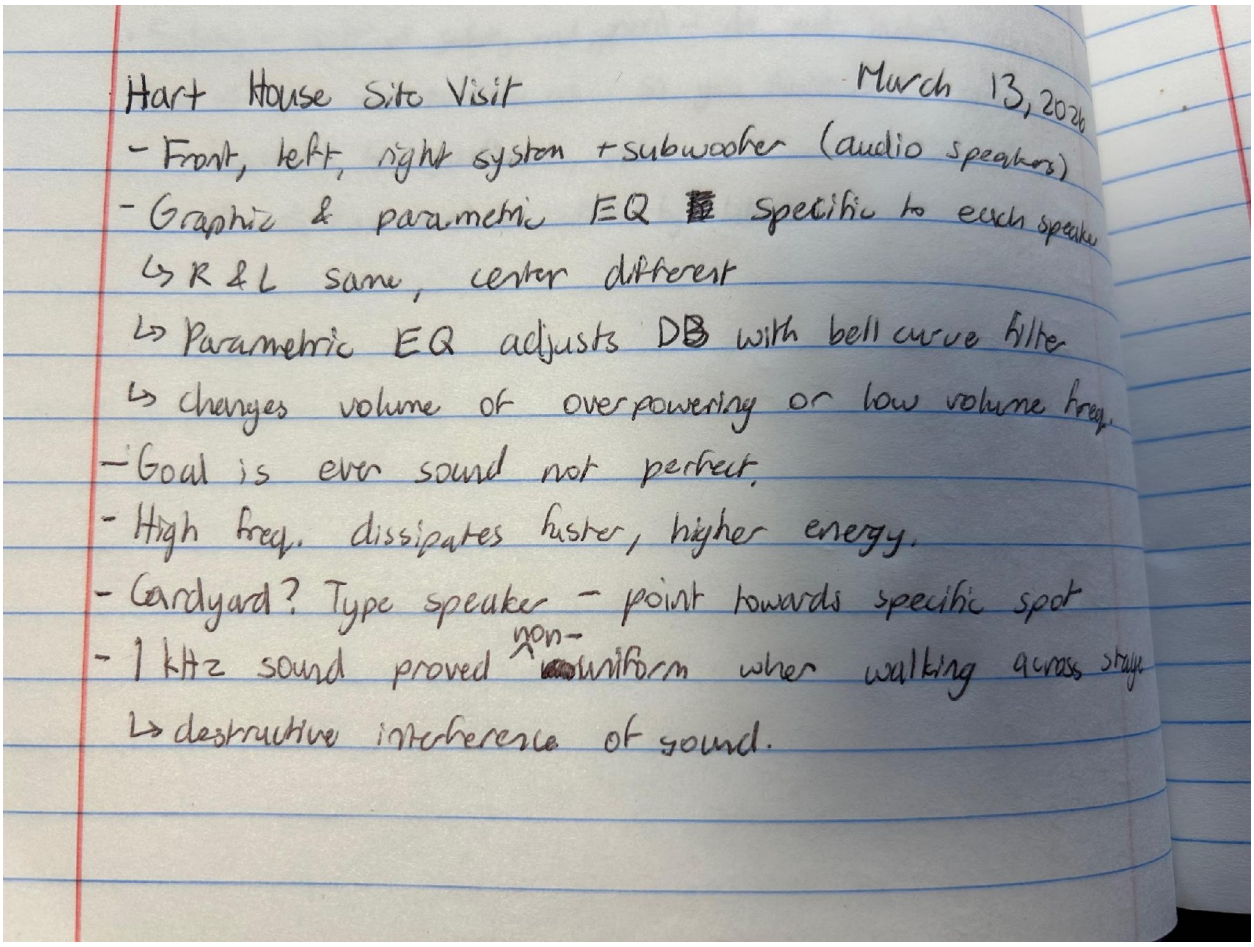


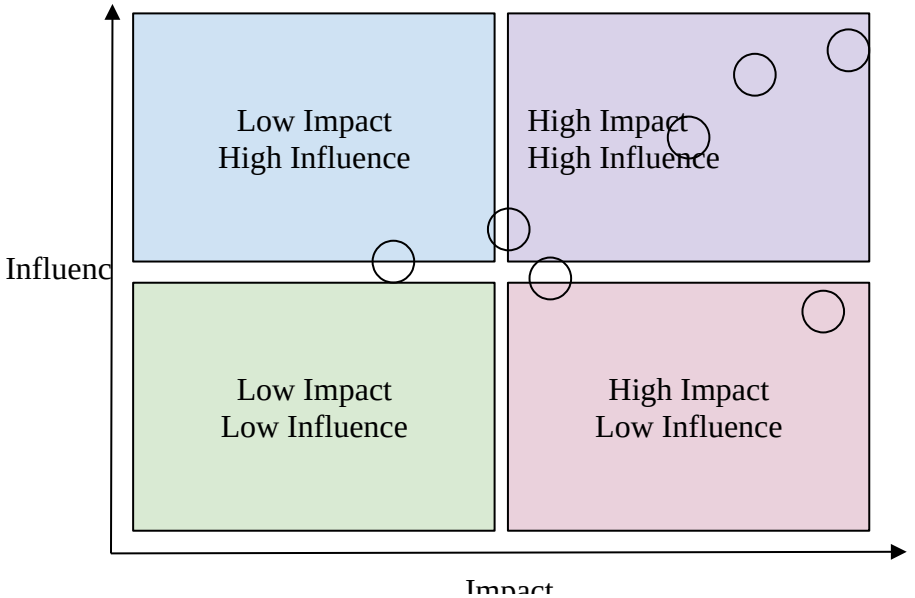
Figure B.4.1. Notes from the site visit

Appendix C: Interest Holders

The 7 interest holders were initially ranked by placing them on an influence-impact matrix and then ranked based on their pairwise comparison scores.

Appendix C.1: Interest Holders Matrix

Determined by their level of impact and influence, the interest holders (following the numbered list) listed on Section 4 are plotted on an Impact vs Influence Matrix to better understand their significance to the project, which is shown in Figure C.1.1



Legend

1 - Dance Performers Using Hart House Theatre

2 - Hart House Theatre Sound

4 - Hart House Stage Managers

5 - Stage Crew of Hart House Theatre

6 - Audience Members of Hart

Figure C.1.1. Impact vs influence of interest holders listed in Section 4.0.

Appendix C.2: Interest Holders Pairwise Comparison

Tables C.2.1 to C.2.3 show the Pairwise Comparison for all Interest holders of the project to help in determining their relevance against each other.

Table C.2-1. Impact Pairwise Comparison for Interest Holders

	1	**2**	**3**	**4**	**5**	**6**	**7**
1		1	1	1	0	0	0
2	0		1	1	0	0	0
3	0	0		0	0	0	0
4	0	0	1		0	0	0
5	1	1	1	1		1	0
6	1	1	1	1	0		0
7	1	1	1	1	1	1	
Total	3	4	6	5	1	2	0

Table C.2-2. Influence Pairwise Comparison for Interest Holders

	1	**2**	**3**	**4**	**5**	**6**	**7**
1		1	1	1	0	0	0
2	0		1	0	0	0	0
3	0	0		0	0	0	0
4	0	1	1		0	0	0
5	1	1	1	1		0	0
6	1	1	1	1	1		1
7	1	1	1	1	1	0	
Total	3	5	6	4	2	0	1

Table C.2-3. Sum of scores of each IH based on Table C.2-1 and Table C.2-2

Interest Holder #	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Impact Score	3	4	6	5	1	2	0
Influence Score	3	5	6	4	2	0	1
Total Score	6	9	12	9	3	2	1
Overall Rank	**4**	**2**	**1**	**3**	**5**	**6**	**7**

Appendix D: Process of Generating Function

To help generate the function of the design, the Black Box Method was used by considering the input and output of the design (Figure D.1.1)

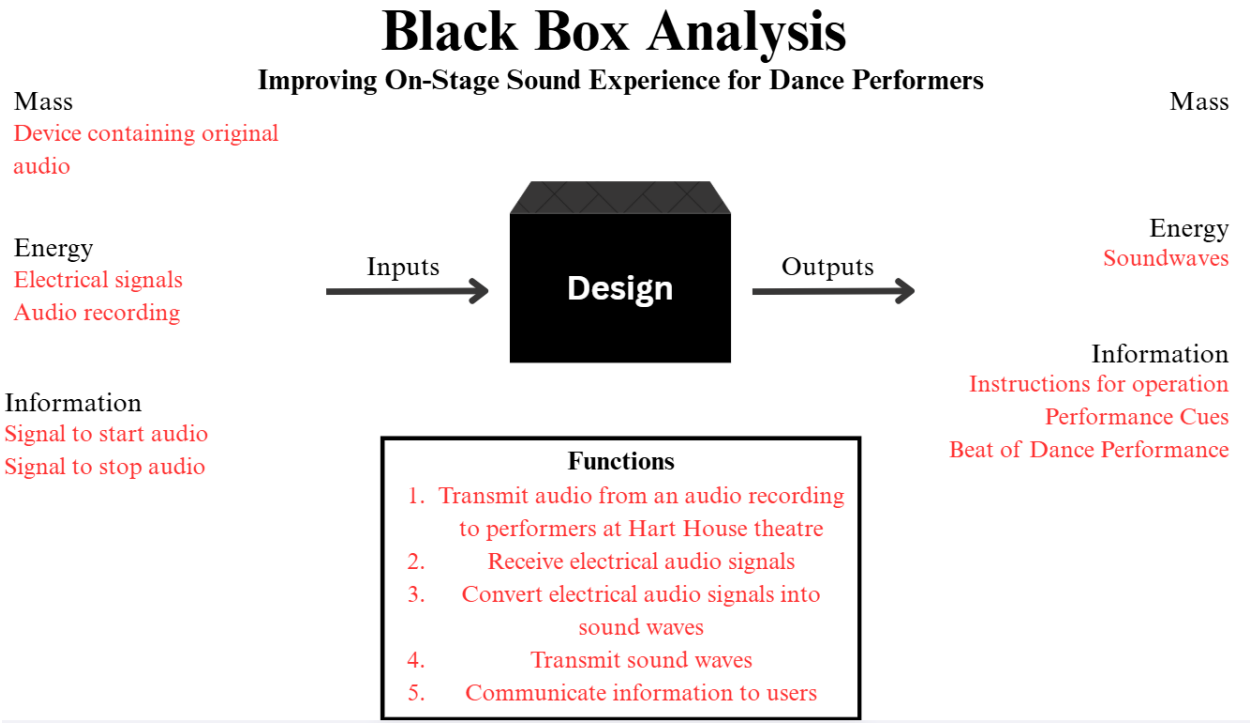


Figure D.1.1. The Black Box Method was used to determine the primary and secondary functions of the design.

Appendix E: Objectives and Constraints

Four objectives and three constraints are listed in Sections 5.2 and 5.3 of the document. This appendix section contains three sub-sections outlining the process of determining the rank of the objectives (via Pairwise Comparison), justifications for each of those objectives, and the constraints justification.

Appendix E.1: Pairwise Comparison

Pairwise comparison is used to help determine the importance of the objectives relative to each other. This tool is shown in Table E.1.1.

Table E.1.1 Pairwise Comparison for Design Objectives

	Affordable	Minimize Audio Latency	Improve Audio Accuracy	Volume of Sound	Ease of Use	**Score**	**Rank**
Affordable		0	0	0	1	1	**4**
Minimize Audio Latency	1		0	1	1	3	**2**
Improve Audio Accuracy	1	1		1	1	4	**1**
Volume of Sound	1	0	0		1	2	**3**
Ease of Use	0	0	0	0		0	**5**

Appendix E.2: Objectives Justification

Table E.2.1 provides justification for the objectives listed in Section 5.2.

Table E.2.1. Objectives Justification

Main Objectives	Specified Metric	Goal	Justification
Improve Audio Accuracy	Frequency Response (dB)	A flat frequency response (+/- 3dB maximum variation) across the frequency range of 20 Hz to 20 kHz. See Appendix E.2 for a sample graph.	• 20Hz - 20kHz frequency range reflects the average hearing range of a person. • A flat frequency response entails that the transmitted sound reproduces the original audio completely unaltered and +/-3dB variation will not be perceivable by the human ears [21].
Minimize Audio Latency	Milliseconds (ms)	≤10 ms	Latency must be at a maximum of 10 milliseconds between individuals to achieve near-instant audio transmittance [22].
Volume on Stage	Decibels (dB)	80-85dB	Ideal long-term listening volume range that prevents hearing damage [23].
Affordable	$CAD	≤$1000 (Appendix A)	Affordable design makes it more easily accessible to all demographics, in line with the client's want (Appendix A).
Ease of Use	Steps to operate	≤3 steps	Benchmarking: In-ear monitors for stage performances are set up in 3 steps [24].

Appendix E.3: Constraints Justification

Table E.2.1 provides justification for the constraints listed in Section 5.3.

Description	Specified Metric	Limit	Justification
Affordability	$CAD	≤$5000	The client stated that the maximum budget is $5000 (Appendix A)
Volume Safety	Decibels (dB)	≤97dB	Assuming performances would not exceed 30 minutes in time duration, the maximum safety threshold for volume is 97 dB [25]
Accuracy of Audio	Frequency Response (dB)	A flat frequency response (+/- 5dB maximum variation) across 20 Hz to 20 kHz.	20Hz - 20kHz frequency range reflects the average hearing range of a person. A frequency response should be ideally flat, meaning that the original sound is transmitted by the source completely unaltered. In practice, this will not be the case. The source states a +/-5dB would be acceptable especially over a large seating area but +/-3dB is preferred [26].
Electrical Safety	Electrical parts left unguarded	0 parts	As per the Ontario Electrical Safety Code (OESC Rules 2-202), covers or guards should remain installed at all times except for maintenance purposes due to the risk of shock [27]

Appendix E.4: Sample Frequency Response Graph

The frequency graph below can be used to understand the process for measuring the success of the design objective outlined in Table 6 of this document.

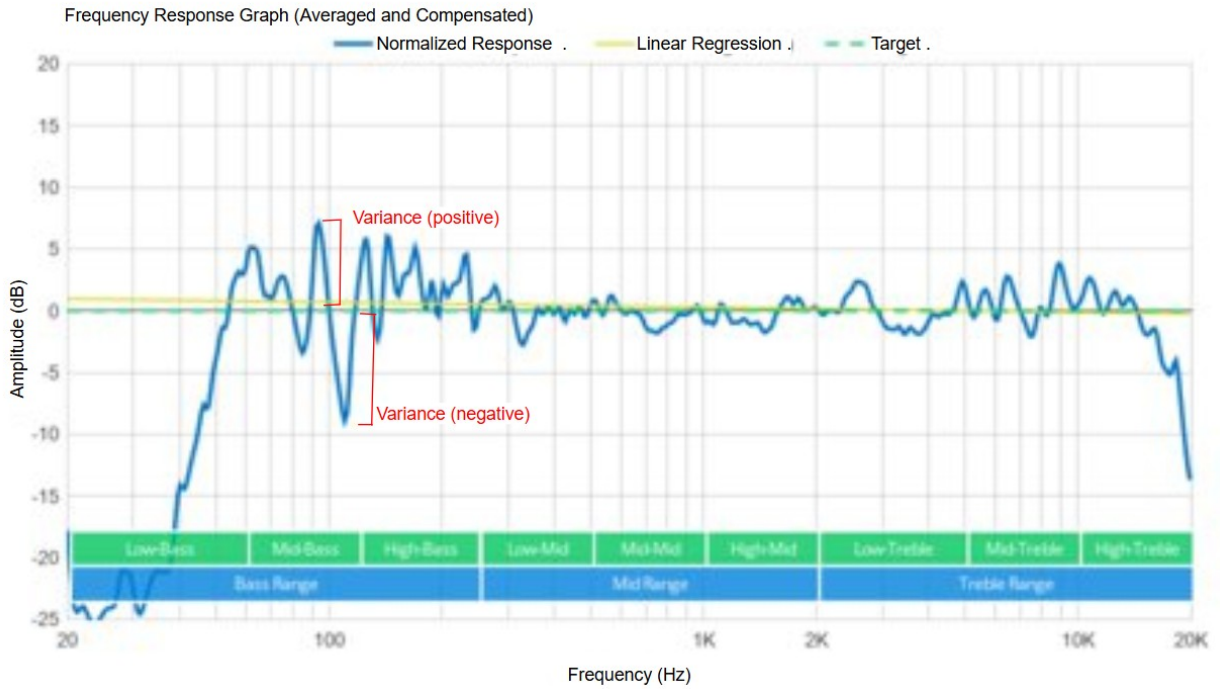


Figure E.2. Frequency Response Graph showing variance from original audio level. (Taken from [39]).

Appendix F: Idea Generation Process

This appendix section contains the sub-sections related to sections 6.0-7.0 in the document above.

Appendix F.1 Brainwriting

This sub-section contains figures of the brainwriting examples used for idea generation.

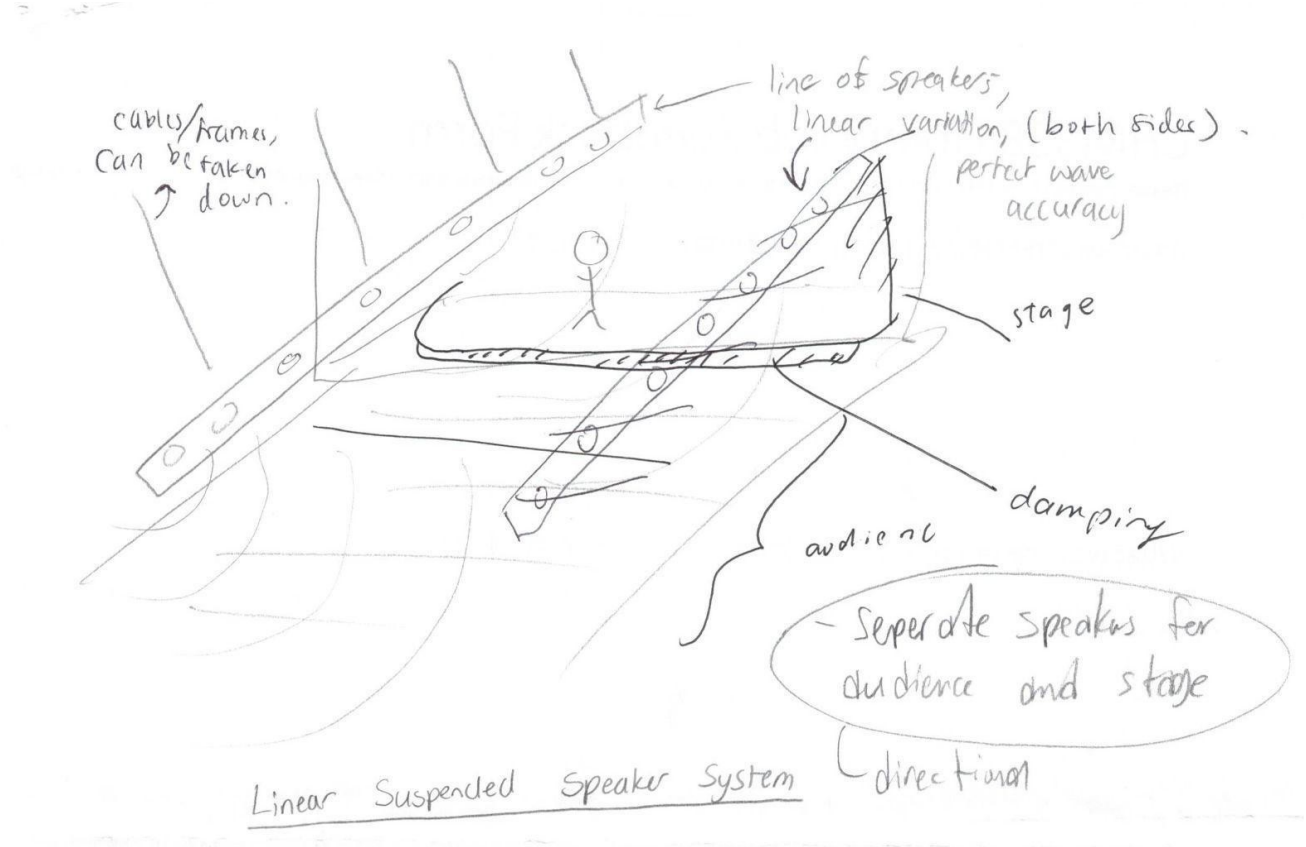


Figure F.1.1. Linear Suspended Speaker System Idea

Figure F.1.2. Dynamic Bass Projection Drone-Based System Idea

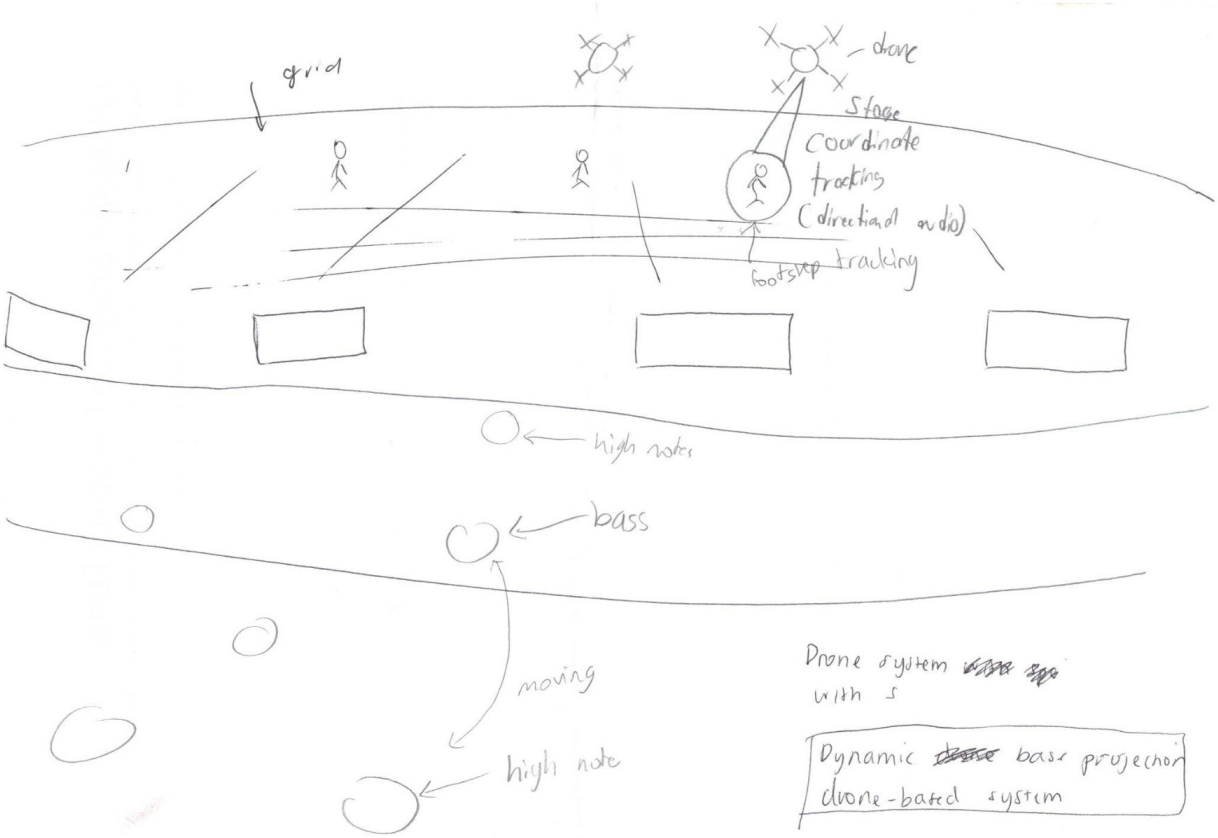


Figure F.1.3. Linear Grounded Speaker System with Live Dynamic Drone Based Audio Correction Idea

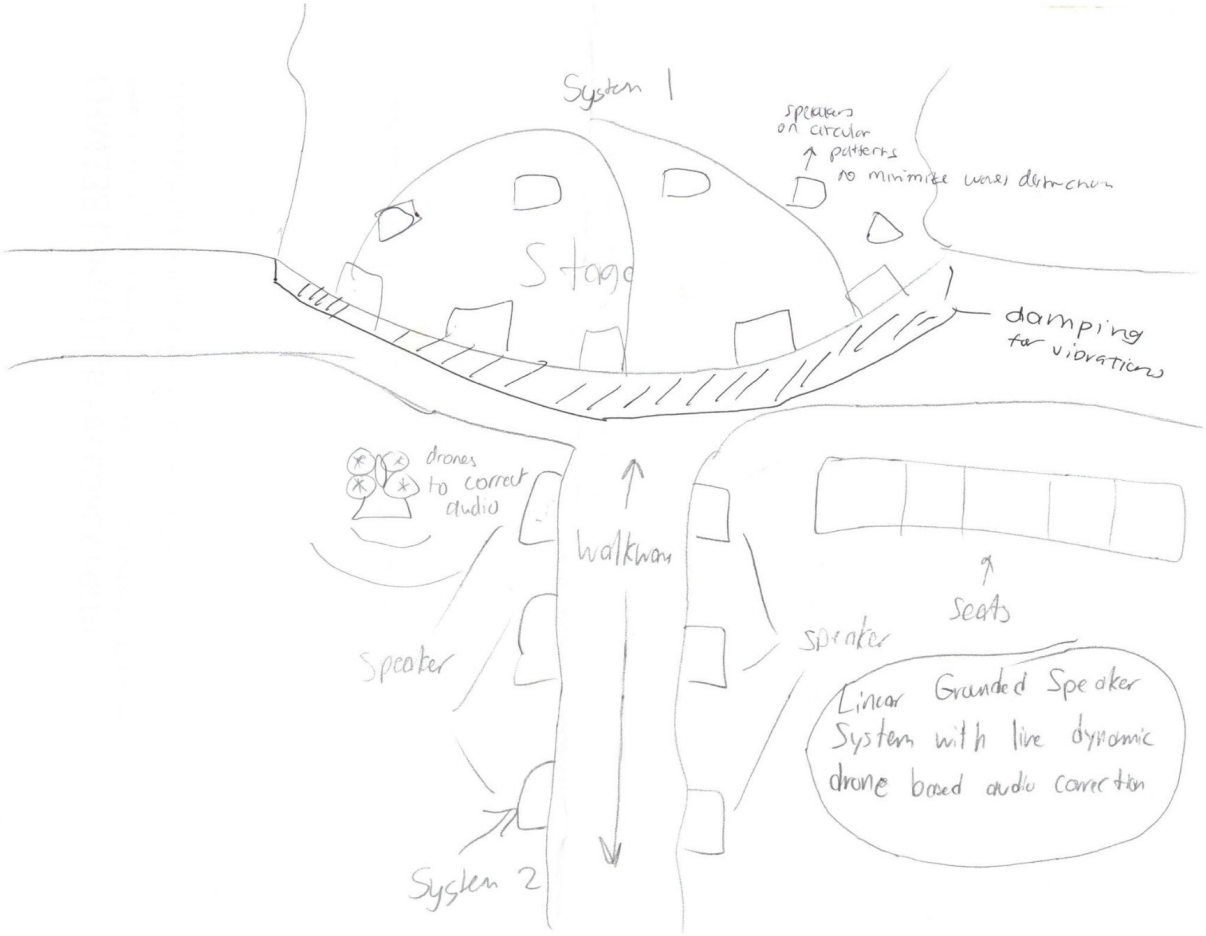
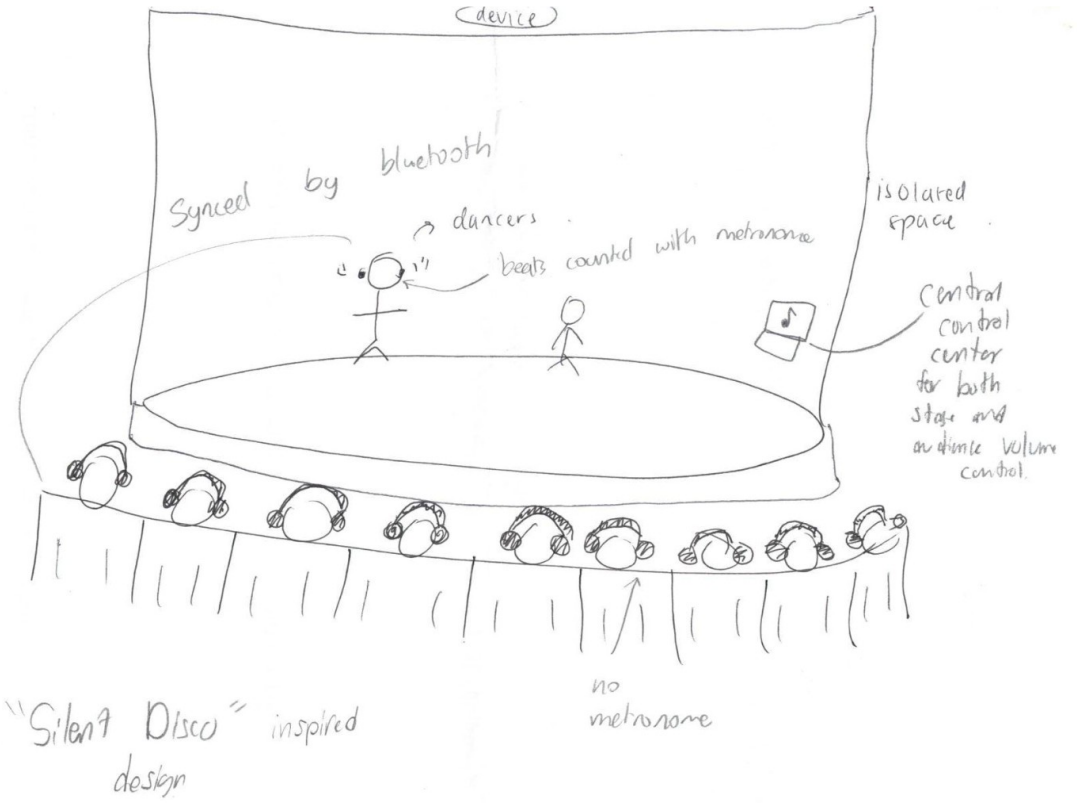


Figure F.1.4. “Silent Disco” Inspired Design Idea



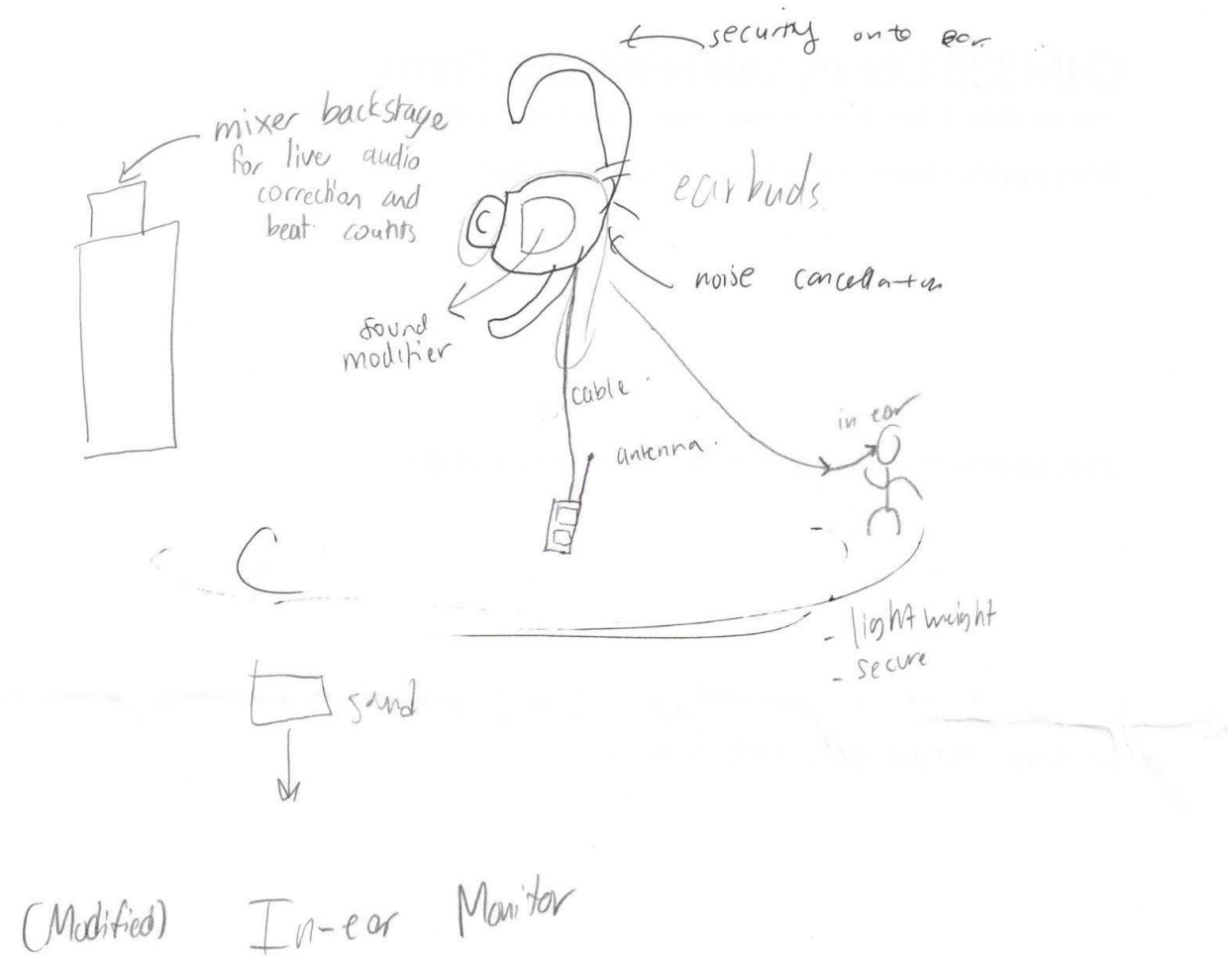


Figure F.1.5. In-Ear Monitor Idea

Appendix F.2: Morph Chart Idea Generation

The Morph Chart tool was used by the team to help generate more solutions. Morph Chart was used since it helps ensure that the solution produced has all the means necessary to achieve all functions and constraints of the design. The Morph Chart is presented in Table F.2.1, and the Morph chart, along with the derived solutions are shown from Table F.2.2 to F.2.6.

Table F.2-1. Morph Chart

Functions, Objectives, Constraints	**Means**	**Means**	**Means**	**Means**	**Means**	**Means**	**Means**
Receive electrical audio signals	Antenna	Bluetooth	Wired				
Convert electrical audio signals into sound waves	Radio signal	Speaker transducer					
Transmit sound waves	Speaker	Earphones	Headphones	Surround sound system	Drone	Singer	Replicate/ Amplify existing sound
Communicate information to users	Paper instructions	Audio instructions	Video instruction	Hired professional assistance			
Improve Audio Accuracy	Align speaker soundwaves	Optimize sound absorption and diffusion	Mixer that reduces high frequency	Mixer that reduces base volume	Optimize room acoustics (walls)		
Minimize Audio Latency	Place the speakers as close as possible	Suspensions from the ceiling	Floor speakers	Sound tunnel			
Volume on Stage	An amplifier for the electrical signals	Reduce the stage and theatre background noise	Damping materials where necessary	Curved panels for amplification			
Electrical Safety	Electrical wall panels	Hide it in the walls	Waterproof materials	Rubber coating	Closed box (Portable)	Electrical tape	

Table F.2-2. Morph Chart Solution #1

Functions, Objectives, Constraints	Means	Means	Means	Means	Means	Means	Means
Receive electrical audio signals	Antenna	Bluetooth	Wired				
Convert electrical audio signals into sound waves	Radio signal	Speaker transducer					
Transmit sound waves	Speaker	Earphones	Headphones	Surround sound system	Drone	Singer	Replicate/ Amplify existing sound
Communicate information to users	Paper instructions	Audio instructions	Video instruction	Hired professional assistance			
Improve Audio Accuracy	Align speaker soundwaves	Optimize sound absorption and diffusion	Mixer that reduces high frequency	Mixer that reduces base volume	Optimize room acoustics (walls)		
Minimize Audio Latency	Place the speakers as close as possible	Suspensions from the ceiling	Floor speakers	Sound tunnel			
Volume on Stage	An amplifier for the electrical signals	Reduce the stage and theatre background noise	Damping materials where necessary	Curved panels for amplification			
Electrical Safety	Electrical wall panels	Hide it in the walls	Waterproof materials	Rubber coating	Closed box (Portable)	Electrical tape	

Solution: Bluetooth speaker system with speakers lined around the stage that replicates and amplifies exisiting sounds

Table F.2-3. Morph Chart Solution #2

Functions, Objectives, Constraints	Means	Means	Means	Means	Means	Means	Means
Receive electrical audio signals	Antenna	Bluetooth	Wired				
Convert electrical audio signals into sound waves	Radio signal	Speaker transducer					
Transmit sound waves	Speaker	Earphones	Headphones	Surround sound system	Drone	Singer	Replicate/ Amplify existing sound
Communicate information to users	Paper instructions	Audio instructions	Video instruction	Hired professional assistance			
Improve Audio Accuracy	Align speaker soundwaves	Optimize sound absorption and diffusion	Mixer that reduces high frequency	Mixer that reduces base volume	Optimize room acoustics (walls)		
Minimize Audio Latency	Place the speakers as close as possible	Suspensions from the ceiling	Floor speakers	Sound tunnel			
Volume on Stage	An amplifier for the electrical signals	Reduce the stage and theatre background noise	Damping materials where necessary	Curved panels for amplification			
Electrical Safety	Electrical wall panels	Hide it in the walls	Waterproof materials	Rubber coating	Closed box (Portable)	Electrical tape	

Solution: A drone with a camera that tracks a dancer's position which is synced with floor speakers that adjusts volume according to the dancer's position.

Table F.2-4. Morph Chart Solution #3

Functions, Objectives, Constraints	Means	Means	Means	Means	Means	Means	Means
Receive electrical audio signals	Antenna	Bluetooth	Wired				
Convert electrical audio signals into sound waves	Radio signal	Speaker transducer					
Transmit sound waves	Speaker	Earphones	Headphones	Surround sound system	Drone	Singer	Replicate/ Amplify existing sound
Communicate information to users	Paper instructions	Audio instructions	Video instruction	Hired professional assistance			
Improve Audio Accuracy	Align speaker soundwaves	Optimize sound absorption and diffusion	Mixer that reduces high frequency	Mixer that reduces base volume	Optimize room acoustics (walls)		
Minimize Audio Latency	Place the speakers as close as possible	Suspensions from the ceiling	Floor speakers	Sound tunnel			
Volume on Stage	An amplifier for the electrical signals	Reduce the stage and theatre background noise	Damping materials where necessary	Curved panels for amplification			
Electrical Safety	Electrical wall panels	Hide it in the walls	Waterproof materials	Rubber coating	Closed box (Portable)	Electrical tape	

Solution: Bluetooth earphones that directly transmit sound to the dancer's ears which reduces the distracting bass and with a slight noise cancelation.

Table F.2-5. Morph Chart Solution #4

Functions, Objectives, Constraints	Means	Means	Means	Means	Means	Means	Means
Receive electrical audio signals	Antenna	Bluetooth	Wired				
Convert electrical audio signals into sound waves	Radio signal	Speaker transducer					
Transmit sound waves	Speaker	Earphones	Headphones	Surround sound system	Drone	Singer	Replicate/ Amplify existing sound
Communicate information to users	Paper instructions	Audio instructions	Video instruction	Hired professional assistance			
Improve Audio Accuracy	Align speaker soundwaves	Optimize sound absorption and diffusion	Mixer that reduces high frequency	Mixer that reduces base volume	Optimize room acoustics (walls)		
Minimize Audio Latency	Place the speakers as close as possible	Suspensions from the ceiling	Floor speakers	Sound tunnel			
Volume on Stage	An amplifier for the electrical signals	Reduce the stage and theatre background noise	Damping materials where necessary	Curved panels for amplification			
Electrical Safety	Electrical wall panels	Hide it in the walls	Waterproof materials	Rubber coating	Closed box (Portable)	Electrical tape	

Solution: Wireless headphones for each audience member to use to listen to the performance with a modified stage environment to redirect and focus sound towards the dancers on stage.

Table F.2-6. Morph Chart Solution #5

Functions, Objectives, Constraints	Means	Means	Means	Means	Means	Means	Means
Receive electrical audio signals	Antenna	Bluetooth	Wired				
Convert electrical audio signals into sound waves	Radio signal	Speaker transducer					
Transmit sound waves	Speaker	Earphones	Headphones	Surround sound system	Drone	Singer	Replicate/ Amplify existing sound
Communicate information to users	Paper instructions	Audio instructions	Video instruction	Hired professional assistance			
Improve Audio Accuracy	Align speaker soundwaves	Optimize sound absorption and diffusion	Mixer that reduces high frequency	Mixer that reduces base volume	Optimize room acoustics (walls)		
Minimize Audio Latency	Place the speakers as close as possible	Suspensions from the ceiling	Floor speakers	Sound tunnel			
Volume on Stage	An amplifier for the electrical signals	Reduce the stage and theatre background noise	Damping materials where necessary	Curved panels for amplification			
Electrical Safety	Electrical wall panels	Hide it in the walls	Waterproof materials	Rubber coating	Closed box (Portable)	Electrical tape	

Solution: Replacement of current sound system with a new modern surround sound system and appropriate site modfications for uniform sound distribution.

Appendix G: Idea Selection Process

This appendix section outlines all methods used to narrow down our list of ideas and means generated from the idea generation process into the final three alternative designs and the final proposed conceptual design. The selection process includes consolidation, feasibility check, multivoting, graphical decision chart, and the Pugh method.

Appendix G.1: Consolidation and Consolidated List of Ideas

The first step of the idea selection is consolidation, which helped the team generate a comprehensive list of full solutions and group them according to the six established themes. This process includes going through each idea, removing duplicates, and combining similar ideas to form a better solution to the design problem. The list of ideas, along with their themes are presented in Table G.1.1.

Table G.1.1. Master list of all ideas sorted by theme

Acoustic/ Architectural Treatment	1. Stage floor and walls to be built with dampening features that minimise vibrations and echoes 2. Damping system under the stage, combined with the existing speaker system 3. Curved panels - install curved reflective panel (similar to satellite dishes) behind the speakers that focuses and redirects sound towards the stage 4. Acoustic lenses - attach acoustic lenses (structured grids) in front of the speakers to ensure sound waves arrive on stage in phase 5. Installation of permanent sound adsorption and diffusers on walls and ceilings to optimize the distribution of sound waves from the existing Hart House speakers 6. Replacing portions of stage floor with vibration absorbing composite panels 7. Vibration mitigator (same concept as noise cancelling where you play the inverse sound wave) 8. Wireless headphones for each audience members to use to listen to the performance, with the addition of a modified sound environment to redirect and focus sound towards the dancers on stage 9. Portable sound absorption and diffusers system
Loudspeaker System Design	1. Total replacement of current sound system with a new modern surround sound system with appropriate site modifications for uniform sound distribution 2. An audio mixer that changes the composition of music to adapt to a specific speaker's frequency variation with speakers, and an online course for use 3. 360° stage sound dome - speakers all around to ensure the sound waves emitted are identical 4. Playing different parts of the audio through different speakers, e.g. stronger bass played from back of theatre 5. A device worn by dancers that, when activated, will create an isolated sound environment for the dancers themselves (like the shields from Dune) 6. Isolated sound bubble, enabling separate sound environment for dancers from

	the audience 7. Isolated sound environments are divided by a soundproof glass wall 8. Speakers system that ensures an aligned soundwave environment 9. Bluetooth speaker system with speakers lined around the stage that replicates and amplifies existing sounds 10. The whole stage floor becomes a speaker 11. Suspended speakers from the ceiling 12. A retractable sound bar at the edge of the stage that projects sounds back towards the dancers 13. Speakers on a string/line lined around the stage 14. Speakers on a conveyor belt (around the stage) that are controlled by the sound operators 15. Speaker on a pendulum (in line with the beat of the song) that are controlled by the sound operators
Proximity Focused	1. Motion-activated speakers which turn on when a dancer is in range to minimize sound delay and sound overlapping 2. Speakers that are placed at the back of the stage for closer proximity to performers 3. Small speaker clips on clothing delivering consistent, small, and minimal audio projection 4. Speakers installed on the same level as the dancers' ears 5. The whole ceiling is a speaker with a specific portion activated only when there is a dancer under it 6. Earpieces/earbuds for all performers & audience members 7. Wearable neck speaker (like the fan device that is worn around the neck) 8. Fabric/wearable antenna that will receive, convert, and transmit sound waves to the dancers 9. Fill the room with helium to make soundwaves coming from the existing speakers travel faster (people wear oxygen masks) 10. Project light to the dancers and make them wear a device that converts light into sound 11. Implanted music chip with an instruction manual to guide user usage prior, during, and after the performance 12. Redesign jewellery and/or prop to become a speaker/to give audio queues with an instruction manual to guide user usage 13. A drone with a camera that tracks a dancer's position which is synced with floor speakers that adjusts volume according to the dancer's position 14. Bone-Conduction Headphones/Earpieces (differs to conventional earpieces by not plugging the ear canal)
Beat/Tactile Focused	1. Bluetooth earpiece with a transmitter that has cues for start and beat 2. Waistband/ankle bracelet/wrist bracelet that delivers vibrations to communicate tempo to the dancers while being a part of the costume (wearable haptic metronome device)

	3. Haptic floor system - floor beats with the music 4. Earpieces for dancers acting solely as metronome 5. Thin, light chest pad that delivers vibrations for tempo 6. Loud metronome system in addition to the existing speaker system 7. Injected nanobots that replicate music beats with an instruction manual to guide user usage prior, during, and after the performance 8. In-ear monitor with dance instructions instead of music 9. An electric pad muscle massager synced to deliver shocks in line with the beat of the song
Concentrated Audio	1. Ultrasonic Directional Speaker System - uses parametric ultrasonic speakers to create a beam of sound aimed at performers 2. LRAD-like device (sound crowd control device used by police) 3. Megaphone 4. Stage monitors/floor speakers that send concentrated/focused sound waves targeting the dancers' ears 5. Speakers on the ceilings that deliver concentrated sound waves that follow the dancers' location/coordinates 6. Very loud speakers to replace the current speakers 7. Make all the audience members use headphones, and the dancers use small, simple speakers as used during rehearsals 8. Bluetooth speakers that directly transmit sound to the dancer's ears which reduces the distracting bass with slight noise cancellation
Visual Cues	1. Attach an angled television to the stage monitors that points towards the performers to communicate visual cues 2. Eye contact with electronics embedded in it to show visual cues synced to the audio transmitted by the existing speakers 3. Colour coded light beams that help to guide dancers synced with the audio from existing speakers 4. Floor screens that project a colour at different vibrancies based on the sound waves of the music, paired with speakers 5. Rayband meta glasses speaker system with holographic projections in the lenses for visual cues

Note: To communicate information to users, each solution will include audio, visual, written instructions and/or professional help for installation.

Appendix G.2: Feasibility Check

The next step in the selection process was conducting a feasibility check. This process includes checking each solution against the functions and constraints, then removing the solutions that does not meet the criteria or modifying them to better fit the functions and constraints. Table G.2.1 presents the feasibility status of each solution, and Table G.2.2 compiles all solutions that passed the feasibility check.

Table G.2.1. Feasibility Check

Themes	#	Idea	Feasibility (Yes/No)
Acoustic/ Architectural Treatment	1	Stage floor and walls to be built with dampening features that minimise vibrations and echoes	Yes
	2	Damping system under the stage, combined with the existing speaker system	Yes
	3	Curved panels - install curved reflective panel (similar to satellite dishes) behind the speakers that focuses and redirects sound towards the stage	Yes
	4	Acoustic lenses - attach acoustic lenses (structured grids) in front of the speakers to ensure sound waves arrive on stage in phase	Yes
	5	Installation of permanent sound adsorption and diffusers on walls and ceilings to optimize the distribution of sound waves from the existing Hart House speakers	Yes
	6	Replacing portions of stage floor with vibration absorbing composite panels	Yes
	7	Vibration mitigator (same concept as noise cancelling where you play the inverse sound wave)	No
	8	Wireless headphones for each audience members to use to listen to the performance, with the addition of a modified sound environment to redirect and focus sound towards the dancers on stage	Yes
	9	Portable sound absorption and diffusers system	Yes
Loudspeaker System Design	10	Total replacement of current sound system with a new modern surround sound system with appropriate site modifications for uniform sound distribution	Yes
	11	An audio mixer that changes the composition of music to adapt to	Yes

		a specific speaker's frequency variation with speakers, and an online course for use.	
	12	360° stage sound dome - speakers all around to ensure the sound waves emitted are identical	No
	13	Playing different parts of the audio through different speakers, e.g. stronger bass played from back of theatre	Yes
	14	A device worn by dancers that, when activated, will create an isolated sound environment for the dancers themselves (like the shields from Dune)	No
	15	Isolated sound bubble, enabling separate sound environment for dancers from the audience	No
	16	Isolated sound environments are divided by a soundproof glass wall	No
	17	Speakers system that ensures an aligned soundwave environment	Yes
	18	Bluetooth speaker system with speakers lined around the stage that replicates and amplifies existing sounds	Yes
	19	The whole stage floor becomes a speaker	No
	20	Suspended speakers from the ceiling	Yes
	21	A retractable sound bar at the edge of the stage that projects sounds back towards the dancers	Yes
	22	Speakers on a string/line lined around the stage	No
	23	Speakers on a conveyor belt (around the stage) that are controlled by the sound operators	Yes
	24	Speaker on a pendulum (in line with the beat of the song) that are controlled by the sound operators	Yes
Proximity Focused	25	Motion-activated speakers which turn on when a dancer is in range to minimize sound delay and sound overlapping	Yes
	26	Speakers that are placed at the back of the stage for closer proximity to performers	Yes
	27	Small speaker clips on clothing delivering consistent, small, and minimal audio projection	Yes

	28	Speakers installed on the same level as the dancers' ears	Yes
	29	The whole ceiling is a speaker with a specific portion activated only when there is a dancer under it	Yes
	30	Earpieces/earbuds for all performers & audience members	Yes
	31	Wearable neck speaker (like the fan device that is worn around the neck)	No
	32	Fabric/wearable antenna that will receive, convert, and transmit sound waves to the dancers	No
	33	Fill the room with helium to make soundwaves coming from the existing speakers travel faster (people wear oxygen masks)	No
	34	Project light to the dancers and make them wear a device that converts light into sound	No
	35	Implanted music chip with an instruction manual to guide user usage prior, during, and after the performance	No
	36	Redesign jewellery and/or prop to become a speaker / to give audio queues with an instruction manual to guide user usage	Yes
	37	A drone with a camera that tracks a dancer's position which is synced with floor speakers that adjusts volume according to the dancer's position	Yes
	38	Bone-Conduction Headphones/Earpieces (differs to conventional earpieces by not plugging the ear canal)	Yes
Beat Focused	39	Bluetooth earpiece with a transmitter that has cues for start and beat.	Yes
	40	Waistband/ankle bracelet/wrist bracelet that delivers vibrations to communicate tempo to the dancers while being a part of the costume (wearable haptic metronome device)	Yes
	41	Haptic floor system - floor beats with the music	Yes
	42	Earpieces for dancers acting solely as metronome	Yes
	43	Thin, light chest pad that delivers vibrations for tempo	Yes
	44	Loud metronome system in addition to the existing speaker system	No
	45	Injected nanobots that replicate music beats with an instruction manual to guide user usage prior, during, and after the	No

		performance	
	46	In-ear monitor with dance instructions instead of music	Yes
	47	An electric pad muscle massager synced to deliver shocks in line with the beat of the song	No
Concentrated Audio	48	Ultrasonic Directional Speaker System - uses parametric ultrasonic speakers to create a beam of sound aimed at performers	Yes
	49	LRAD-like device (sound crowd control device used by police)	No
	50	Megaphone	No
	51	Stage monitors/floor speakers that send concentrated/focused sound waves targeting the dancers' ears	Yes
	52	Speakers on the ceilings that deliver concentrated sound waves that follow the dancers' location/coordinates	Yes
	53	Very loud speakers to replace the current speakers	No
	54	Make all the audience members use headphones, and the dancers use small, simple speakers as used during rehearsals	Yes
	55	Bluetooth speakers that directly transmit sound to the dancer's ears which reduces the distracting bass with slight noise cancellation	Yes
Visual Cues	56	Attach an angled television to the stage monitors that points towards the performers to communicate visual cues	Yes
	57	Eye contact with electronics embedded in it to show visual cues synced to the audio transmitted by the existing speakers	No
	58	Colour coded light beams that help to guide dancers synced with the audio from existing speakers	Yes
	59	Floor screens that project a colour at different vibrancies based on the sound waves of the music, paired with speakers	No
	60	Rayband meta glasses speaker system with holographic projections in the lenses for visual cues	No

Table G.2.2. Solutions That Passed the Feasibility Check

Total # of solutions	**Idea**
Acoustic/Architectural Treatment	
8	• Stage floor and walls to be built with dampening features that minimise vibrations

	and echoes ● Damping system under the stage, combined with the existing speaker system ● Curved panels - install curved reflective panel (similar to satellite dishes) behind the speakers that focuses and redirects sound towards the stage ● Acoustic lenses - attach acoustic lenses (structured grids) in front of the speakers to ensure sound waves arrive on stage in phase ● Installation of permanent sound adsorption and diffusers on walls and ceilings to optimize the distribution of sound waves from the existing Hart House speakers ● Replacing portions of stage floor with vibration absorbing composite panels ● Wireless headphones for each audience members to use to listen to the performance, with the addition of a modified sound environment to redirect and focus sound towards the dancers on stage ● Portable sound absorption and diffusers system
Loudspeaker System Design	
9	● Total replacement of current sound system with a new modern surround sound system with appropriate site modifications for uniform sound distribution ● An audio mixer that changes the composition of music to adapt to a specific speaker's frequency variation with speakers, and an online course for use ● Playing different parts of the audio through different speakers, e.g. stronger bass played from back of theatre ● Speakers system that ensures an aligned soundwave environment ● Bluetooth speaker system with speakers lined around the stage that replicates and amplifies existing sounds ● Suspended speakers from the ceiling ● A retractable sound bar at the edge of the stage that projects sounds back towards the dancers ● Speakers on a conveyor belt (around the stage) that are controlled by the sound operators ● Speaker on a pendulum (in line with the beat of the song) that are controlled by the sound operators
Proximity Focused	
9	● Motion-activated speakers which turn on when a dancer is in range to minimize sound delay and sound overlapping ● Speakers that are placed at the back of the stage for closer proximity to performers ● Small speaker clips on clothing delivering consistent, small, and minimal audio projection ● Speakers installed on the same level as the dancers' ears ● The whole ceiling is a speaker with a specific portion activated only when there is a dancer under it ● Earpieces/earbuds for all performers & audience members

	• Redesign jewellery and/or prop to become a speaker/to give audio queues with an instruction manual to guide user usage • A drone with a camera that tracks a dancer's position which is synced with floor speakers that adjusts volume according to the dancer's position • Bone-Conduction Headphones/Earpieces (differs to conventional earpieces by not plugging the ear canal)
Beat Focused	
6	• Bluetooth earpiece with a transmitter that has cues for start and beat • Waistband/ankle bracelet/wrist bracelet that delivers vibrations to communicate tempo to the dancers while being a part of the costume (wearable haptic metronome device) • Haptic floor system - floor beats with the music • Earpieces for dancers acting solely as metronome • Thin, light chest pad that delivers vibrations for tempo • In-ear monitor with dance instructions instead of music
Concentrated Audio	
5	• Ultrasonic Directional Speaker System - uses parametric ultrasonic speakers to create a beam of sound aimed at performers • Stage monitors/floor speakers that send concentrated/focused sound waves targeting the dancers' ears • Speakers on the ceilings that deliver concentrated sound waves that follow the dancers' location/coordinates • Make all the audience members use headphones, and the dancers use small, simple speakers as used during rehearsals • Bluetooth speakers that directly transmit sound to the dancer's ears which reduces the distracting bass with slight noise cancellation
Visual Cues	
2	• Attach an angled television to the stage monitors that points towards the performers to communicate visual cues • Colour coded light beams that help to guide dancers synced with the audio from existing speakers

Note: To communicate information to users, each solution will include audio, visual, written instructions and/or professional help for installation.

Appendix G.3: Multi-Voting

Multi-voting was used as a structured prioritization method to narrow down our list of solutions in Table G.3. Each team member was allocated four votes, which they could freely distribute across different preferred solutions based on the project's detailed requirements, with specific emphasis on objectives. Votes were then tallied, and all solutions that received no votes were removed to ensure collective judgement within the team. This left the team with 12 solutions:

1. Suspended speakers from the ceiling
2. Bluetooth earpiece with a transmitter that has cues for start and beat
3. Redesign jewellery and/or prop to become a speaker/to give audio queues with an instruction manual to guide user usage
4. Stage floor and walls to be built with dampening features that minimize vibrations and echoes
5. An audio mixer that changes the composition of music to adapt to a specific speaker's frequency variation with speakers, and an online course for use
6. Speakers system that ensures an aligned soundwave environment
7. Make all the audience members use headphones, and the dancers use small, simple speakers as used during rehearsals
8. Attach an angled television to the stage monitors that points towards the performers to communicate visual cues
9. Playing different parts of the audio through different speakers, e.g. stronger bass played from back of theatre
10. Total replacement of current sound system with a new modern surround sound system with appropriate site modifications for uniform sound distribution
11. Speakers on the ceilings that deliver concentrated sound waves that follow the dancers' location/coordinates
12. Portable sound absorption and diffusers system

Upon further discussion, the team decided to apply the SCAMPER method to combine several solutions to better achieve the objectives of the project. This left the team with 8 solutions which are presented in Table G.3.1.

Table G.3.1. Final 8 solutions after the multi-voting process and applying SCAMPER method

# Combined/old #	New #	Idea
1 + 10 + 5	**1**	Suspended modern surround sound system with appropriate site modifications for uniform sound distribution (with audio mixer)
2 + 3	**2**	Redesign jewellery to become a bluetooth earpiece with a transmitter that has cues for start and beat
4 + 8	**3**	Attach an angled television to the stage monitors, combined with dampening features that minimise vibrations and echoes
6	**4**	Speakers system that ensures an aligned soundwave environment

7	**5**	Make all the audience members use headphones, and the dancers use small, simple speakers as used during rehearsals
9	**6**	Playing different parts of the audio through different speakers, e.g. stronger bass played from back of theatre
11	**7**	Speakers on the ceilings that deliver concentrated sound waves that follow the dancers' location/coordinates
12	**8**	Portable sound absorption and diffusers system

Appendix G.4: Graphical Decision Chart

The graphical decision chart was used to compare our final 10 voted solutions against our top 2 objectives to arrive at our final 3 alternative design solutions. The top two objectives were determined via Pairwise Comparison, as discussed in Appendix E.1. The positioning of each solution on the chart was done by team discussion.

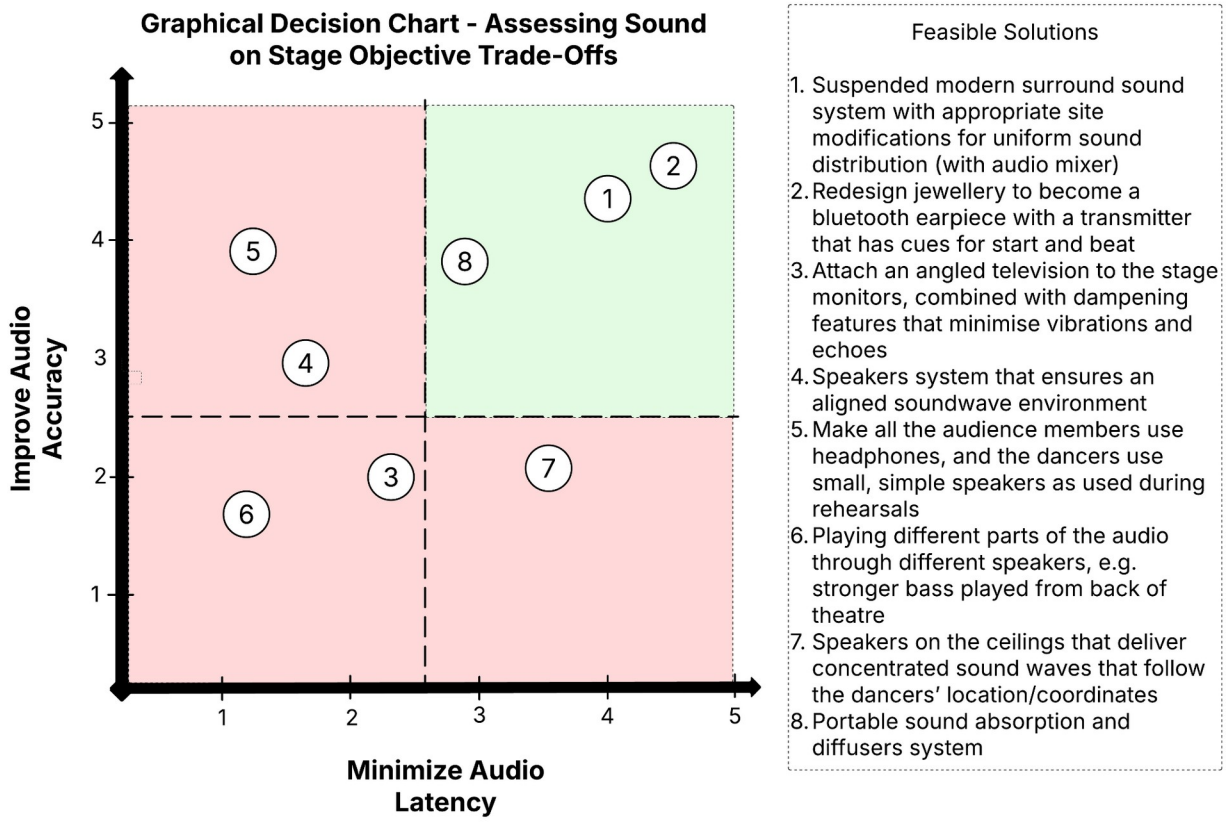


Figure G.4. Graphical Decision Chart

Appendix H: Immersed-On-Stage: Detailed Design Description

To provide a more comprehensive review of Immersed-On-Stage, this appendix section provides detailed specifications of our first alternative design, as presented in Table H.1.

Table H.1. Detailed Specifications of Immersed-On-Stage

Specification	Description	Justification
Components	● 5.1 Surround Speaker System ○ Front L/C/R (3) - Kilpsch RP-640D ○ 1 pair of left and right surround speakers (2) - Kilpsch RP-140D ○ 1 subwoofer - Kilpsch RP-121SW ● Suspension System ○ Ceiling Speaker Mount (6) - MM-3RDX-18 Indoor Ceiling Mount	Our most important objective is to improve audio accuracy. The speakers cited are high quality theatre-grade speakers used for surround sound audio systems. See section 6.3.1 for how surround systems improve on this objective. Professional installation is required. See all specifications cited below.
Kilpsch RP-640D	Weight: 1.7 kg Height: 28.5” Width: 6” Depth: 2.4” Frequency Response: 55Hz-25kHz +/- 3dB [40]	Information will be helpful for suspension and accurate placement of speakers. The +/- 3dB variation range is well within our objective.
Kilpsch RP-140D	Weight: 1.7 kg Height: 12” Width: 6” Depth: 2.4” Frequency Response: 69Hz-25kHz +/- 3dB [41]	Information will be helpful for suspension and accurate placement of speakers. The +/- 3dB variation range is well within our objective.
Kilpsch RP-121SW	Weight: 16 kg Height: 16.6” Width: 16” Depth: 19.7” Frequency Response: 28-120 Hz [42]	Information will be helpful for suspension and accurate placement of speakers. The subwoofer only covers the lower end of frequency ranges or bass.
MM-3RDX-18 Indoor Ceiling	Adjustable Ceiling Mount with 3 degrees of freedom (180	- Degrees of freedom allow speakers to be angled for optimal sound accuracy.

Mount	degree tilt) ● Supports up to 34.1kg [43]	- Weights of speakers fall within the weight constraint of the mount.
Cost Analysis	Kilpsch 140D (2) - $349.00CAD ea. [44] Kilpsch 640D (3) - $849.00CAD ea. [45] Kilpsch R-121SW (1) - $545.00 CAD ea. [46] MM-3RDX-18 (6) - $202.40 CAD ea. [47] Total: $4968.40 CAD	Cost analysis shows the total meets the constraint for the project but not the objective. The design is still valid and will be considered.

Below are the research and calculations for the placement and angles of the 7 speakers outlined above:

Stage dimensions are taken from the service environment section of the document (Section 3)

- Length varies from 29’4” to 36’9”
- Width: 39’6”
- Usable length: 39’9” due to backstage area
- Usable width: 21’4” due to curved wall and curtain line
- l = 12.12m
- w = 6.5m

Assume length and width boundaries are walls. Please see the diagram below for the standard angles referenced below.

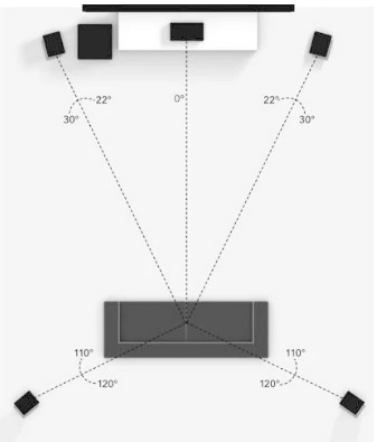


Figure H.1. Standard angles for a 5-speaker surround sound system. (Taken from [48]).

Listening Position based on source [49] : 38% of room width = .38 * 6.5 = 2.47m

Remaining back width = 4.03m
Half of stage length = 12.12 / 2 = 6.06m

Centre speaker: directly at front and center: 6.06m from wall

Front Left Speaker: Using the range of 22-30 degree angles from the vertical for front left and right speakers in a 5 speaker layout, we will be using **25 degrees** as the angle from the vertical line extending from the centralized listening position.
Distance from center = 2.47 tan 25° = 1.15m
Distance from left wall = 6.06 - 1.15 = **4.91m from left wall**

Front Right Speaker: Same as left, but **4.91m from right wall**
Front Left and Right are spaced 2.30m apart.

Back Left Speaker: Using the range of 60-70 degree angles from the vertical for back left and right speakers in a 7 speaker layout, we will be using **65 degrees** as the angle from the vertical line extending from the centralized listening position.
The speaker will be along the side walls rather than the back. (25 degrees from horizontal instead)
Distance from 2.47m mark = 6.06 tan 35° = 2.82m
Distance from back wall = 4.03 - 2.82 = **1.20m from back wall**

Back Right Speaker: Same as left, **1.20m from back wall**
Back Left and Right are spaced 2.40m apart.

Subwoofer: Can be experimented with in terms of placement, but the front left or right corner would be an ideal place to start. [50]

Appendix I: AcoustiFlex: Detailed Design Description

To provide a more comprehensive review of AcoustiFlex, this appendix section provides detailed specifications of our third alternative design, as presented in Table I.1.

Table I.1. Detailed Specifications of AcoustiFlex

Specification	Description	Justification
Components	● 12 Broadband Acoustic Panels ● 4 Quadratic Diffusers	Our client expressed that one of the most prevalent problems on stage is the overpowering low frequency (Appendix A), which was also confirmed during our site visit (Figure …). Therefore, the existence of an absorber would help solve low to mid frequency buildup by absorbing the sound waves energy, and the presence of diffusers help in preventing over absorption of sound to maintain the "liveliness" of the sound, which otherwise would cause the room to have an anechoic chamber quality. The absorber and diffuser would be designed as two separate units to maximize its purpose in treating the stage acoustics. A one-unit acting as both is considered ineffective since it would not possess enough characteristics of both a single absorber and diffuser. The provision of 12 acoustic panels and 4 diffusers would ensure enough coverage on the rear and back portion of a medium sized theatre like Hart House Theatre.
Absorber Dimensions	Length: 2' or 0.6 m Height: 6' or 1.8 m Thickness: 4" or 100mm Edge treatment: ¼ top bottom round	The selection of panel's dimensions are based upon Hart House Stage dimensions as specified in Section 3.1 (Figure …). The thickness of the absorber panels ensures adequate and effective absorption of low to mid frequencies [50]. The introduction of edge treatment to the panels helps reduce edge diffraction, which is described as the bending and bouncing of soundwaves in every direction [51]. Without edge treatment, the rebounded waves would

		cause interference, phase cancellations, and ultimately, irregularities in frequency response [51]
Absorber Material Specification	Mineral wool covered with specialized polyester	Mineral wool is an industry standard and is better compared to other materials, such as fiberglass and foam, in absorbing low to mid range frequency due to its denser quality [52]. The use of specialized polyester for coverage ensures that the cover is acoustically transparent to allow sound to pass through freely.
Noise Reduction Coefficient (NRC) of the Absorbing Panels	0.95 - 1.05	Mineral wool has a high NRC rating, ranging from 0.95 - 1.05 [53]
Diffuser Dimensions	Length: 2’ or 0.6 m Height: 6’ or 1.8 m Thickness: 13.44’ or 34 cm Edge treatment: ¼ top bottom round	The length and height dimensions are similar to the absorber panels to match it visually. The thickness of the panels is based on the maximum well depth of the diffuser.
Diffuser Material Specification	Softwood, such as cherry wood	Wood is an industry standard material for diffusers and scatters back a sound with a nice and balanced tonal quality [54].
Diffuser Type	● Quadratic diffuser ● Root Sequence: 11. ● Maximum well depth: 13.44 inches or 34 cm ● Well width: 1.9 inches or 4.8 cm ● Fin width: 0.25 inches or 0.64 cm	The type of diffuser used in quadratic diffuser with a root sequence of 11. The well depths are varied to target ranges of frequencies, which are calculated using quadratic residue values [55]. The well maximum depth dictates the lowest frequency scattered (~250 Hz) (quarter wavelength rule), while the well width dictates the highest frequency targeted (~3500 Hz) (half wavelength rule) [56].
Well depths	The depths of the first to the last wells from left to right are 0, 1.49, 5.97, 13.44, 7.47, 4.48, 4.48, 7.47, 13.44, 5.97, and 1.49 inches, respectively.	Measurements are calculated using quadratic residue values [55]

Frame	Hardwood material	Ensure frame durability
Base	Steel T base	Provide tip resistance, ensuring the ability to withstand loads from the panel.
Casters	4 lockable casters attached to the steel T base	Ensure reliable lock on stage floor when used during performance.

The panels, both absorption and diffusing panels, should be placed correctly to ensure it contributes to improving the room's acoustic quality. The recommended placements are based on two things, which are [57]:

1. Placing them in the opposite location to the speakers
2. Placing them in the first reflection points

The first reflection point is defined as the location on the surface (walls) where sound waves first get reflected before reaching the listeners' ears. Figure I.1 provides a useful visual that helps in defining the concept of first reflection points.

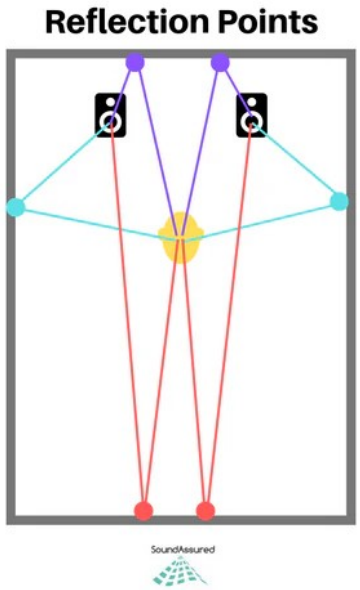


Figure I.1. First reflection points are indicated by the dots (purple, cyan, and red), while the lines represent sound waves coming from the speakers [58]

These placement rules affect the panels placement on the Hart House Theatre Stage, which are shown in the document main body. [59]

Appendix J: Constraint Assessment

This section provides a comprehensive assessment of each alternative design presented in Section 6.3 against the design constraints as outlined in Section 5.3. Table J.1 to J.3 shows the assessment of Immersed-On-Stage, AuricleSync, and AcoustiFlex, respectively.

Table J.1. Immersed-On-Stage Constraint Assessment

Constraint	Assessment
Affordability	Estimates of the design's total cost indicate the possibility of exceeding the constraint.
Volume Safety	Improved sound distribution reduces localized loud spots, helping maintain sound levels below the ≤97 dB safety threshold when properly calibrated.
Accuracy of Audio	The design supports the ±5 dB frequency response constraint by minimizing interference and ensuring consistent sound delivery, though performance depends on precise speaker placement and tuning.
Electrical Safety	• Proper insulation, secure mounting, and adherence to electrical safety standards. • Inaccessible to unauthorized users.

Table J.2. AuricleSync Objective Assessment

Constraint	Assessment
Affordability	Cost constraints limit the choice of materials and wireless technology, requiring a balance between performance quality and economic feasibility. Hence, leveraging on an existing part of the performance allows the design to be kept within the budget.
Volume Safety	Direct audio delivery to the ear introduces the risk of excessive sound exposure. Volume-limiting mechanisms can be further incorporated to ensure safe listening levels and prevent hearing damage for the performers.
Accuracy of Audio	Maintaining consistent audio quality across multiple devices may be challenging due to signal interference or hardware variability. Calibration and robust transmission protocols are required as part of the design's testing phase.
Electrical Safety	The device must be insulated and designed to prevent overheating or short circuits, especially due to close contact with the skin during a performance. The wireless feature ensures safety for the performers with minimal presence of external electrical features exposed.

Table J.3. AcoustiFlex Objective Assessment

Constraint	Assessment
Affordability	Benchmarking: Companies, such as GIK Acoustics, market a 60 * 120 cm acoustic panel for $152 [32], and a 50*117 cm Quadratic Diffuser for $626 [33]. Effective production and scaling strategies can help ensure compliance with the cost constraints (<$5000).
Volume Safety	By minimizing constructive sound interference and enhancing audio intelligibility, the design indirectly supports compliance with the volume safety constraint (≤97 dB) due to reduced need for high volume levels.
Accuracy of Audio	Contribution from both panels significantly improves audio reproduction by minimizing resonance effects and preventing interferences. This results in a more uniform wave distribution across the stage, reducing deviations from the original audio signal.
Electrical Safety	The panels are not equipped with any electrical components; hence, do not possess any electrical hazards.

Appendix K: Pugh Chart - Proposed Conceptual Design Selection

To aid in deciding the final proposed conceptual design, Pugh Chart was used, where design 2 (AuricleSync) and design 3 (AcoustiFlex) were compared against the first design (Immersed-On-Stage) as a baseline (datum). The evaluation used a 5-point scale system with +2 represents a much better quality and +1 represents better quality, and the same applies to -2 and -1, while 0 represents the same as baseline. The criteria defined on the left-most column is the objectives established on Section 5.2. The Pugh Chart is presented in Table K.1.

Table K.1. Pugh Chart

	Alternative Designs				
Objective	Datum: Immersed-On-Stage	Auricle Sync	Justification	Acousti Flex	Justification
Improve Audio Accuracy	S	+2	Compared to Design 1, AuricleSync delivers audio directly to the ear, eliminating room acoustics effects like reflections and interference, resulting in more consistent and accurate sound	0	AcoustiFlex improves accuracy by reducing reflections and standing waves, but it still depends on the existing PA system and cannot fully eliminate acoustic distortions
Minimize Audio Latency	S	+1	AuricleSync reduces latency differences across performers by bypassing sound propagation delays and delivering synchronized wireless audio directly.	-1	AcoustiFlex does not directly reduce latency since it does not modify signal transmission, only slightly improving perceived timing through reduced reverberation.
Volume on Stage	S	+1	AuricleSync reduces reliance on high-volume speakers, resulting in more controlled and consistent sound levels across performers.	0	AcoustiFlex improves sound clarity, allowing lower volume levels to achieve the same audibility, although it still depends on speaker output, which this is significantly improved in design one
Affordable	S	+2	AuricleSync uses smaller, scalable	0	AcoustiFlex avoids structural installation costs

			components and avoids expensive installation, making it significantly more cost-effective.		but may still become expensive depending on the number of panels and diffusers required.
Ease of Use	S	+1	AuricleSync is easier to use as it requires minimal setup (wear and connect), unlike the professional installation required for suspended speakers.	+1	AcoustiFlex requires manual placement and adjustment of panels before each use, but still less complex than the datum.
Total	0	7		0	x

Appendix L: Measures of Success Data Collection Table

Frequency Response testing data collection will involve the measurement of the volume levels of different frequencies or frequency ranges. Table L.1 shows a sample chart of the type of data collection involved in our process.

Table L.1. Data Collection for MoS Frequency Response Test.

Frequency Range	AuricleSync Maximum Frequency Variation (dB)	Hart House Maximum Frequency Variation (dB)	Original Audio Maximum Frequency Variation (dB)	Variation difference between Original and Auricle Sync	Objective Satisfied? (Y/N)
Bass (20 - 250 Hz)					
Low Midrange (250 - 500 Hz)					
Midrange (500 Hz - 2 kHz)					
Upper Midrange (2 - 4 kHz)					
Presence and Brilliance (4 - 16 kHz)					